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ORIGINAL ARTICLE

Special Issue: Touchscreen Testing to Investigate the Neurochemistry of Cognition

The longitudinal behavioral effects of acute exposure to galactic cosmic radiation in female C57BL/6J mice: Implications for deep space missions, female crews, and potential antioxidant countermeasures

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Abstract

Galactic cosmic radiation (GCR) is an unavoidable risk to astronauts that may affect mission success. Male rodents exposed to 33-beam-GCR (33-GCR) show short-term cognitive deficits but reports on female rodents and long-term assessment are lacking. We asked: What are the longitudinal behavioral effects of 33-GCR on female mice? Also, can an antioxidant/anti-inflammatory compound (CDDO-EA) mitigate the impact of 33-GCR? Mature (6-month-old) C57BL/6J female mice received CDDO-EA (400 μg/g of food) or a control diet (vehicle, Veh) for 5 days and Sham-irradiation (IRR) or whole-body 33-GCR (0.75Gy) on the 4th day. Three-months post-IRR, mice underwent two touchscreen-platform tests: (1) location discrimination reversal (tests behavior pattern separation and cognitive flexibility, abilities reliant on the dentate gyrus) and (2) stimulus–response learning/extinction. Mice then underwent arena-based behavior tests (e.g. open field, 3-chamber social interaction). At the experiment's end (14.25-month post-IRR), an index relevant to neurogenesis was quantified (doublecortin-immunoreactive [DCX+] dentate gyrus immature neurons). Female mice exposed to Veh/Sham vs. Veh/33-GCR had similar pattern separation (% correct to 1st reversal). There were two effects of diet: CDDO-EA/Sham and CDDO-EA/33-GCR mice had better pattern separation vs. their respective control groups

Abbreviations: 33-beam-GCR, 33-beam galactic cosmic radiation; 3-CSI, 3-chamber social interaction; AALAC, Association for Assessment and Accreditation of Laboratory Animal Care; Acq, acquisition; BLAF, Brookhaven Laboratory Animal Facility; BNL, Brookhaven National Laboratory; CDDO-EA, 2-cyano-3, 12-dioxooleana-1, 9-dien-28-oic acid-ethylamide; CNS, central nervous system; DCX, doublecortin; DCX+, doublecortin-immunoreactive; EPM, elevated plus maze; Ext, extinction; GCR, galactic cosmic radiation; h, hour(s); Hab, habituation; IACUC, Institutional Animal Care and Use Committee; IHC, immunohistochemistry; IRR, irradiation; LDR Test, location discrimination reversal testing; LDR Train, location discrimination reversal training; LDR, location discrimination reversal; min, minute(s); mon, month(s); MT, must touch; NASA, National Aeronautics and Space Administration; NOR, novel object recognition; NSRL, NASA Space Radiation Laboratory; OF, open field; PFC, prefrontal cortex; PI, punish incorrect; ROUT, robust outlier testing; S−, non-rewarded stimulus; s, second(s); S+, rewarded stimulus; SGZ, subgranular zone; Veh, vehicle.

S. Yun and F.C. Kiffer contributed equally as first authors. G.L. Bancroft, C.S. Guzman, and I. Soler contributed equally as second authors.

[Correction added on 18 January 2025 after first online publication: The Special Issue title has been corrected.]

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(Veh/Sham, Veh/33-GCR), and CDDO-EA/33-GCR mice had better cognitive flexibility (reversal number) vs. Veh/33-GCR mice. One radiation effect/CDDO-EA counter-effect also emerged: Veh/33-GCR mice had slower stimulus–response learning (days to completion) vs. all other groups, including CDDO-EA/33-GCR mice. In general, all mice showed normal anxiety-like behavior, exploration, and habituation to novel environments. There was also a change relevant to neurogenesis: Veh/33-GCR mice had fewer DCX+ dentate gyrus immature neurons vs. Veh/Sham mice. Our study implies space radiation is a risk to a female crew's longitudinal mission-relevant cognitive processes and CDDO-EA is a potential dietary countermeasure for space-radiation CNS risks.

KEYWORDS
arena-based classical rodent behavioral tests, CDDO-EA, hippocampus, Mars-relevant, space radiation, touchscreen operant behavior

1 | INTRODUCTION

The interplanetary space between Earth and Mars consists of a complex field of dynamic particle radiation, derived primarily from galactic cosmic rays (GCR). GCR can penetrate current spacecraft shielding, and thus the risk of GCR to humans on interplanetary missions merits thorough evaluation (Nelson, 2016). GCR risk estimates rely on animal models exposed to accelerated particles at energies encountered in space (Schimmerling, 2016). However, particle accelerators have been limited in the number of particle types and energies that can be generated in a reasonable time-frame (Simonsen et al., 2020). Therefore, most space radiation risk estimates have relied on studies exposing animals to monoenergetic single particle types whose energies, on average, are higher than the true multi-energetic, multi-particle spectrum GCR encountered in space. Fortunately, GCR production capabilities at Brookhaven National Laboratory are now sufficiently advanced that scientists can assess if exposure to whole-body multi-particle irradiation (IRR)—including 33-ion GCR (33-GCR)—alters cognition when compared to sham IRR. Taken together, studies show single-ion, multi-ion, and even 33-GCR disturb a range of rodent central nervous system (CNS) functions, including memory, pattern separation, anxiety, vigilance, social novelty, and motor control (Cekanaviciute et al., 2018; Clément et al., 2020; Kiffer et al. 2019b). However, narrowing these studies to just those that use multi-particle GCR (multi-ion, up to 33-GCR) highlights some contradictory results (Alaghband et al., 2023; Boutros et al., 2021; Britten et al., 2022; Kiffer et al., 2018; Kiffer et al., 2020; Kiffer et al., 2021; Klein et al., 2021; Krukowski et al. 2018a; Krukowski et al., 2021; Puukila et al., 2023; Raber et al., 2016; Raber et al., 2019; Raber et al., 2020; Raber et al., 2021; Schaeffer et al., 2022). In addition, while most work has been performed in just male rodents, some of these studies show sex-specific outcomes of multi-ion whole-body exposure. Thus, the impact of multi-ion GCR, particularly the Mars-relevant 33-GCR, on social and higher cognitive function in female mice warrants more research. In addition, while a few papers have now looked at the behavioral impact of 33-GCR up to 6-months (mon) post-IRR (Alaghband et al., 2023; Desai et al., 2023), more work is needed to assess the even longer-term impact of 33-GCR, particularly given concerns about astronaut morbidity.

In addition to studying the sex-specific and long-term impact of GCR exposure, there is a need to either prevent or ameliorate the CNS effects of GCR. Prevention strategies—such as improved or additional spacecraft shielding—are challenging. In fact, additional shielding may paradoxically lead to absorption of higher GCR dose because of particle fragmentation, back-scatter, and linear energy transfer modulation (Naito et al., 2021; Warden & Bayazitoglu, 2021). Because of this, amelioration strategies—such as the use of pharmacological countermeasures—have received significant attention (DiCarlo et al., 2022; Dynan et al., 2022; Singh & Seed, 2022). Medical countermeasures for space radiation exposure are currently evaluated by lifetime carcinogenic mortality (Werneth et al., 2022); no guidelines or modeling exist for the mitigation of acute, in-flight CNS or operational performance risks for flight crews (Nelson et al., 2016). However, the ideal pharmacological countermeasure for in-flight use would have the following characteristics: (1) high distribution to target organs; (2) high biological half-life; (3) amenable to non-invasive administration; (4) be radioprotective and mitigative, acting via multiple points of radiogenic damage, including DNA damage, oxidative stress, and inflammation, and promote tissue repair (Pariset et al., 2021). With these criteria in mind, 2-cyano-3, 12-dioxooleana-1, 9-dien-28-oic acid-ethylamide (CDDO-EA) warrants careful examination as a potential countermeasure. CDDO-EA is a successful pharmacotherapeutic in animal models for diseases that target the CNS and other organs, acting by promoting endogenous antioxidant pathways, protecting and promoting DNA repair, reducing inflammation, and promoting cellular repair (Crowley et al., 2017; Lei et al., 2020; Luitel et al., 2020; Mathis & Cui, 2016; Neymotin et al., 2011; Stack et al., 2010; Wang et al., 2014; Wible et al., 2018). The human version of CDDO-EA known as omaveloxolone (Skyclarys, CDDO-ME) was recently approved by the FDA

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for the treatment of Friedreich's ataxia, a neuromuscular disorder (Biogen.com). Recently, we reported that male mice exposed 6-mon prior to 33-GCR and concomitantly given CDDO-EA had worse social novelty than sham-irradiated mice, and that CDDO-EA was not an effective countermeasure (Kiffer et al., 2021). However, the impact of 33-GCR and the countermeasure potential of CDDO-EA on female mice and their CNS function are unknown.

There were four goals to this study. First, we wanted to define the impact of 33-GCR on female cognitive ability with relevance to mission success. We used the translationally-relevant touchscreen platform to probe multiple domains of cognitive performance in female mice previously exposed to 33-GCR or sham IRR, the potential countermeasure CDDO-EA chow or a control diet, and a combination of these factors. Our choice of touchscreen task was guided by past work showing IRR changed function linked to the integrity select brain regions. For example, hippocampal dysfunction is seen in a sex- and task-specific manner up to 8-mon post-^{56}Fe IRR (Soler et al., 2021; Whoolery et al., 2020) and 2–3.5 months post-33-GCR (Alaghband et al., 2023). Absent from this literature is the impact of 33-GCR on the performance of female mice in behavioral pattern separation, an ability that is assessed by the touchscreen test location discrimination reversal (LDR; McTighe et al., 2009; Oomen et al., 2013). Dysfunction in the prefrontal cortex (PFC) is also evident in the IRR literature, including IRR-induced dysfunction in cognitive flexibility in female and male rodents (Bellone et al., 2015; Britten et al., 2014; Britten et al., 2022; Britten et al., 2023; Burket et al., 2021; Desai et al., 2023; Hadley et al., 2015; Rudobeck et al., 2017; Shukitt-Hale et al., 2000; Soler et al., 2021; Villasana et al., 2013; Whoolery et al., 2020). However, there is no work examining the impact of 33-GCR on the performance in two PFC-involved touchscreen tasks: LDR (Soler et al., 2021; Swan et al., 2014; Yun et al., 2023), specifically the reversal which is reliant on PFC, and extinction of simple stimulus–response learning (Mar et al., 2013). Notably, the acquisition phase of the simple stimulus–response learning task relies on two other brain regions: the striatum and amygdala (Fernando et al., 2013). Performance on the simple stimulus–response learning task has not yet been assessed after exposure to 33-GCR. It would be important to do this since older work shows single-particle exposure disrupts amygdala and striatal functional integrity (Kiffer et al. 2019b). There is an additional benefit to reaching this first goal via touchscreen testing: Traditional studies in radiation have, with a few exceptions, relied on arena-based tests that involve animals freely-behaving in open areas. These arena-based tests provide excellent insight into behavior, but the open spaces also are anxiogenic in prey species, such as rodents. Arena-based also typically involve only a few trials, limiting temporal resolution of behavioral data captured. The touchscreen-based platform involves sound-isolated, confined spaces with large, versatile visual stimuli and the ability to have appetitive rewards for correct stimulus interaction, prompting repeated, high-resolution behavioral evaluations that are highly adaptive and can probe multiple brain systems. This leads to our second goal: probe multiple brain circuit integrity by testing both on the touchscreen platform (via LDR and simple stimulus–response learning/extinction) and on a battery of arena-based tests. In addition to probing the function of multiple brain regions after 33-GCR exposure, our third goal was to assess the ability of CDDO-EA to prevent 33-GCR-induced changes in long-term behavior or cognition. Prior work has shown CDDO-EA given before, during, and after 33-GCR or sham IRR blunted GCR-induced changes in social behavior when assessed 6-mon later (Kiffer et al., 2021). However, the radioprotective impact of CDDO-EA in female mice has not yet been assessed. Our final goal was to evaluate behavior over many months and conclude with a meaningful cellular measure. To this end, once the months of behavior and cognitive testing in these female mice were complete, we quantified a gold standard index relevant to adult hippocampal neurogenesis: dentate gyrus doublecortin-immunoreactive (DCX+) cells. Altered dentate gyrus neurogenesis indices, such as DCX+ cell number, correlate with changes in many behavioral domains, ranging from memory to spatial learning to the social behavior (DeCarolis et al., 2014; Rivera et al., 2013; Whoolery et al., 2017; Whoolery et al., 2020; Zanni et al., 2018). Decreased neurogenesis indices are seen after whole-body exposure to single particle types at Mars-mission-relative doses, and some, but certainly not all, work suggests that the number of dentate gyrus DCX+ immature neurons correlates with hippocampal function, such as improved pattern separation. However, to-date it is unknown how indices relevant to neurogenesis change 14-mon after any type of particle irradiation, and certainly not 14-mon after 33-GCR.

With these four goals in mind, we set out to understand the potential risks of 33-GCR to the CNS. We exposed mature (6-mon-old) female C57BL/6J mice to the whole-body Mars mission-relevant dose of 75 cGy. Subgroups of Sham and 33-GCR mice were given CDDO-EA or Vehicle (Veh) chow before, during, and immediately after IRR, allowing us also to assess CDDO-EA as a candidate countermeasure in astronaut-aged female mice. This CDDO-EA (or Veh) administration scheme allowed this antioxidant to reach saturated plasma concentration at the time of IRR and remain high through the acute post-IRR period when potential for IRR-oxidative damage is high (Kim et al., 2012; Wang et al., 2014). Over the subsequent 13.25 mon post-IRR, we assessed spaceflight-relevant behavioral and cognitive domains using both operant touchscreen and arena-based testing. We also examined the cellular outcome of an index of hippocampal neurogenesis in Sham versus 33-GCR mice at 14.25 mon post-IRR.

2 | METHODS

2.1 | Animals and housing

Six mon-old female C57BL/6J mice (a total of 64 mice, Jackson Laboratory, Bar Harbor, ME; RRID:IMSR_JAX:000664) were ordered and shipped directly to Brookhaven National Laboratory (BNL), specifically the Brookhaven Laboratory Animal Facility (BLAF). The day before IRR, mice were transported in their same cages from the BLAF to the NSRL (NASA Space Radiation Laboratory) and were returned

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to BLAF the day after IRR. At BLAF and NSRL, mice were housed 4/cage in a HEPA-filtered, closed airflow vivarium system under a 12:12 h light/dark cycle (06:00 light on) at 22°C, 30%–70% humidity and given a standard rodent chow (Standard Rodent Chow 5015; Lab Diet, cat. no. 0001328) and water ad libitum. Two days post-IRR, mice were transported to Children's Hospital of Philadelphia (CHOP) by ground shipment. At CHOP, mice were held in quarantine for 1.5 mon (6 weeks) with ad libitum access to custom-made chow (containing 13 PPM ivermectin and 150 PPM fenbendazole to prevent ectoparasites and nematode endoparasites; Test Diet, custom cat. no. 1813527[5SKU]). Behavioral testing began at 2.5 mon (10 weeks) post-IRR when mice were released from quarantine and were back on regular chow (Standard Rodent Chow 5015). At CHOP, mice were housed in a HEPA-filtered, closed airflow vivarium system (Lab Products Inc., cat. no. Enviro-Gard™ A) under a 12:12 h light/dark cycle (06:15 light on) at 20–23°C and 30%–40% humidity and given a standard rodent chow (Standard Rodent Chow 5015) and water ad libitum. At cage change, each cage was given a nestlet square; otherwise, no enrichment was provided. All care and procedures were approved by the Institutional Animal Care and Use Committees (IACUC) at BNL and CHOP and held in accordance with the Association for Assessment and Accreditation of Laboratory Animal Care (AAALAC) and National Institute of Health (NIH) guidelines for the care and use of laboratory animals (CHOP, Philadelphia, PA; AAALAC Accreditation #000427, PHS Animal Welfare Assurance D16-00280 [OLAW A3442-01]) and BNL (Upton NY; AAALAC Accreditation #000048, PHS Animal Welfare Assurance D16-00067 [OLAW A3106-01]). Our scientific reporting adheres to the ARRIVE 2.0 guidelines (Percie du Sert et al., 2020).

2.2 | Diet

After 3 days of acclimation to the BNL BLAF, mice were split into 2 diet groups (Veh or CDDO-EA) based on pre-assignment (Section 2.10). Mice received either Veh (Lab Diet cat. no. 5002 mixed with 1.25% EtOH, 3.75% Neobee Oil) or CDDO-EA chow (Veh chow with 0.02% RTA 405; Reata Pharmaceuticals, cat. no. RTA 405) ad libitum for 5 days (Kim et al., 2012). Both Veh and CDDO-EA chow were prepared by Purina Mills, LLC. The IRR (Section 2.3) took place on Day 4 of the 5-day diet.

2.3 | Irradiation (IRR)

On Day 4 of the Veh or CDDO-EA diet, mice in both Diet groups were split into Sham or 33-GCR groups based on pre-assignment (See Section 2.10; $n=16$ per Diet/Radiation group; Figure 1). The IRR with Sham or 33-GCR took place during the BNL 19A campaign, as previously described (Kiffer et al., 2021). In brief, IRR mice in each cage (2–3 mice/cage) were placed together in well-ventilated 10×10×4.5 cm holders. They were then exposed to 75 cGy of NASA's whole-body 33-beam 33-GCR in a 60×60 cm beam spread.

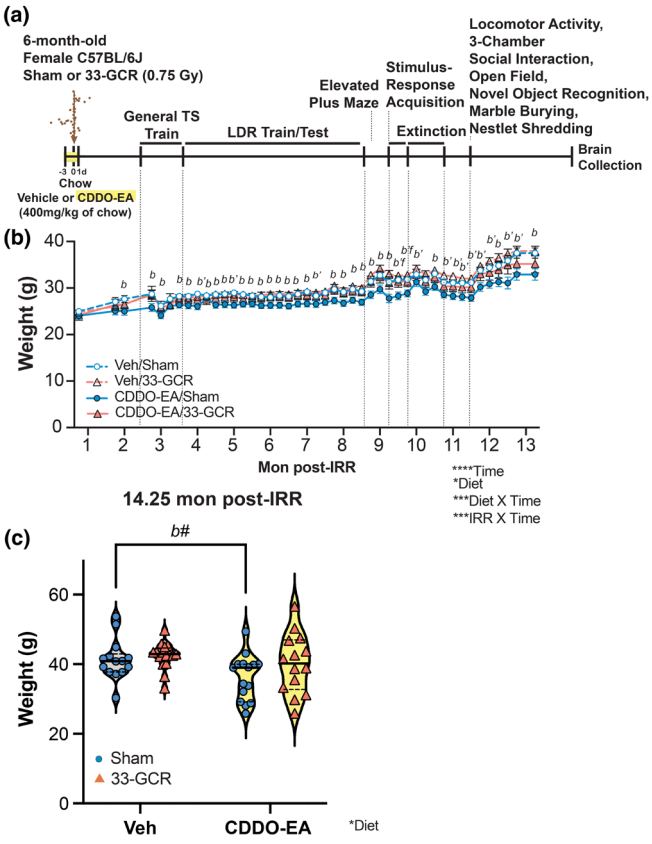


FIGURE 1 Female C57BL/6J mice given the antioxidant/anti-inflammatory agent CDDO-EA chow and exposed to Sham IRR weighed slightly less than Veh/Sham mice from 2 month post-IRR onward. This lower weight was not seen in CDDO-EA/33-GCR mice. (a) Timeline of experiment. Female mice from Jackson Laboratory were delivered to Brookhaven National Laboratory (BNL) at 6-months (mon) of age. After acclimation to the BNL animal housing facilities, mice were given CDDO-EA (400 mg/kg of food) in chow or vehicle chow (Veh) for 5 days. On Day 4 of CDDO-EA or Veh diet, mice were exposed to one of two IRR conditions: Whole-body 33-GCR (0.75 Gy) or Sham IRR. Mice were then shipped to Children's Hospital of Philadelphia Research Institute where they underwent general training on a touchscreen (TS) platform followed by a task to assess behavioral pattern separation, Location Discrimination Reversal (LDR) and a simple stimulus–response learning and extinction learning. Mice then went through a series of classical arena-based behavioral tests prior to brain collection at 14.25 mon. (b) Body weight of mice over the course of experiment. (c) Weight at 14.25 mon. Statistical analysis in B: Mixed effected three-way RM ANOVA (IRR×Diet×Time); in C: Two-way ANOVA (IRR×Diet). Main effect and/or interaction are denoted by $^{*}p<0.05$, $^{**}p<0.01$, $^{***}p<0.001$, $^{****}p<0.0001$; post-hoc multiple comparison analysis significance denotations between groups: Veh/Sham versus CDDO-EA/Sham, b $p<0.05$, b' $p<0.01$, $b\#=0.065$; CDDO-EA/Sham versus CDDO-EA/33-GCR, f $p<0.05$. 14–16 mice per group. Dotted vertical lines indicate the point on the behavioral timeline where weight was recorded in B and violin plots depict median (solid line) and 25% and 75% quartiles (dotted line) in (c). Complete subject numbers and detailed data analysis are provided in Table S1.

The 75 cGy was delivered over ca. 1.5 h, and beam uniformity and dosimetry were evaluated and monitored by the staff at NSRL. The remaining mice that received Sham exposure were also paired with

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cagemates in the holders for 1.5 h but were not placed on the beam line. The four experimental groups that resulted were Veh/Sham, Veh/33-GCR, CDDO-EA/Sham, and CDDO-EA/33-GCR.

2.4 | General physiological measures

Mouse body weight data were regularly collected from their arrival at BNL at 6-mon of age through behavioral testing and until brain collection 14.25 mon (57 weeks) post-IRR. "Survival" of mice was recorded throughout the experiment, with the survival graph reflecting mice that were lost because of possible diet and/or radiation effects; mice that were lost because of husbandry issues were not considered in the survival graph (see Section 2.10). Alopecia (coat state) scores were recorded twice (10.5 mon or 42 weeks post-IRR and 13.5 mon or 54 weeks post-IRR) as general indication of stress or maladaptation to treatment (Bechard et al., 2011).

2.5 | Behavioral testing

2.5.1 | Overview of behavioral testing

Two and one-half mon (10 weeks) post-IRR, mice began behavioral testing (Figure 1a). Mice first underwent touchscreen training and testing interspersed with and followed by arena-based tests to assess early to late effects of GCR on multiple behavioral domains, as described in separate sections below. Mice were weighed regularly to ensure that weights did not fall below 80% of their starting weight. Mice that exceeded this weight loss limit would be excluded from the experiment; however no mouse met this exclusion criterion.

Mice underwent touchscreen training and testing in the Bussey-Saksida operant touchscreen platform (Lafayette Life Sciences, cat. no. 80614A) used with the ABET Core Intelli-Interface for Easy Multi-Chamber Control (Lafayette Life Sciences, cat. no. 81430) and ABET II Software for Touch Screens with Audio-Video Stimulus (Lafayette Life Sciences, cat. no. 89509). From 2.5 to 8.6 mon post-IRR, mice underwent general touchscreen training followed by location discrimination reversal training (LDR Train; program LD1 choice reversal v3, Lafayette Life Sciences, cat. no. 89546-6) and location discrimination reversal testing (LDR Test) to assess behavioral pattern separation and cognitive flexibility. From 9.25 to 10.8 mon post-IRR, the same mice underwent Acquisition training (Acq) and then Extinction testing (Ext; Lafayette Life Sciences, cat. no. 89547). During training and testing on the touchscreen platform, mice were food restricted Monday through Friday, with food only provided 4 h each day after touchscreen testing. Food was provided ad libitum from Friday afternoon through Sunday evening. Data were automatically recorded from the operant touchscreen platform.

Mice underwent arena-based testing from 8.6 to 8.9 mon post-IRR (elevated plus maze [EPM]) and from 10.8 to 12 mon post-IRR (locomotor activity, 3-chamber social interaction [3-CSI], open field [OF], novel object recognition [NOR], marble burying, nestlet shredding). During these time periods, mice were given unrestricted food access. For arena-based testing (with the exception of marble burying and nestlet shredding), data and video recordings were captured by a ceiling-mounted Ace camera (Basler, cat. no. acA640–90gc) and tracking was performed by Ethovision XT (Noldus Information Technology, RRID:SCR_000441). Nose point, center point, and tail-base were captured. Nose points were used for exploratory measures, whereas center points were used for gross locomotion. Marble burying was scored by the handler performing the experiment, and nestlet shredding was assessed by weighing pre- and post-test intact materials.

In all tests, the handler placed the test mouse in the testing apparatus and left the room for the duration of the trial. Between mice, behavioral equipment was disinfected and deodorized with 10% ethanol or 10% Birex SE III Concentrate (Biotrol, cat. no. 96043) and allowed to dry. The four experimental groups were run through tests in a counterbalanced order to minimize the effect of testing order.

Touchscreen training and testing

Touchscreen testing was performed between 08:00 and 14:00 daily (Monday through Friday). The rewarding appetitive stimulus ("reward") used throughout training and testing was Ensure® Original Strawberry Nutrition Shake (Abbott Laboratories, RRID:SCR_010477).

2.5.2 | General touchscreen training and food restriction overview

Prior to behavioral testing in the, mice moved through several general training stages, as previously described (Horner et al., 2013; Soler et al., 2021; Whoolery et al., 2020). Specifically, general touchscreen training consists of six phases, two of which are set for one-day duration (Habituation [Hab] 1 and Hab 2) and four of which are criterion-based (and thus have varied durations): Initial Touch, Must Touch (MT), Must Initiate, and Punish Incorrect (PI).

Hab 1 (1 day) began 2.5 mon post-IRR and consisted of 10 min of free movement in the touchscreen chamber. A 2 × 6 window grid was provided, but no reward, tone, or house light was provided. This was the first time the mice were in the touchscreen chamber. Beginning 14 h prior to Hab 2 and continuing for the duration of touchscreen experiments, mice were restricted to their standard chow beginning at 18:00 each evening until 14:00 the following day during touchscreen training or testing sessions. From Friday (14:00) to Sunday (17:00), unrestricted food was provided. Mice were weighed regularly to ensure that weights did not fall below 80% of their starting weight. Mice that exceeded this weight loss limit would be excluded from the experiment, however no mouse met this exclusion criterion.

For Hab 2 (1 day), mice were placed in the same touchscreen chambers for 20 min. The reward tray light was on, and an initial priming reward of 150 μL was provided simultaneously with a tone (70 dB, 500 Hz, 1 s). After the mouse inserted and withdrew their

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head from the reward tray, the tray light turned off, there was a 10 s delay, followed by reactivation of the tray light and tone, and 7 μL of Ensure was dispensed into the reward tray.

For the subsequent Initial Touch phase (1–2 days, criteria 25 trials/30 min), mice began with one of the bottom six grids pseudorandomly lit. The stimulus was not displayed in the same position more than 3 times in a row. Mice had 30 s to touch the lit square. If the correct square was touched, 21 μL of the rewarding stimulus (three times the usual reward) was dispensed, the reward tray lit up and a tone was played. If, however, there was an omission (the correct square was not touched) by 30 s, the reward tray dispensed only 7 μL of the reward stimulus; the tray still was illuminated and a tone still played. Regardless of response, there was a 10 s intertrial interval (ITI) where the reward tray light turned off. To complete this phase, mice must complete 25 trials (regardless of touch or omission) in 30 min. Mice that completed 25 trials were promptly removed from the arena and returned to their home cage.

Next, mice learned MT (criteria 25 trials/30 min), where they must touch the screen stimuli to receive the appetitive reward. This phase was identical to the Initial Touch phase but required mice to touch the correct lit square on the screen to receive the reward. Correct touches resulted in only 7 μL of the reward in conjunction with the reward tray light and tone. Incorrect touches (touches to unlit squares) yield no reward or light or sound cues.

Once MT criteria were reached, mice then advanced to the Must Initiate phase (criteria 25 trials/30 min), where trials do not begin until a mouse inserts their head into the reward tray. Reward rules and completion criteria were the same as in the MT, but once reward was collected, mice must reinsert their head into the reward tray to initiate the following trial.

Once mice reached the Must Initiate criteria, they moved onto the final training phase, PI (criteria: total trial counter 30, consecutive 76% correct for the first day and 80% correct for the 2nd day in a row). This phase "punished" incorrect touches (touching an unlit grid square) by turning on the chamber light for 5 s. Then a correction trial occurred, where the same rewarding stimulus square as the previous trial was once again lit. A correction trial was repeated until the mouse touched the rewarding stimulus. Correction trials do not count in the total trial counter in regard to completion criteria.

2.5.3 | Location discrimination reversal training (LDR train) and testing (LDR test)

After general touchscreen training, mice then proceeded to LDR training (LDR Train [50 Max trials/30 min] and testing (LDR Test [81 Max trials, 8 Max reversal]), program LD1 choice reversal v3; Lafayette Life Sciences, cat. no. 89546-6). In the LDR Train phase, mice learned that touching one of two squares separated by an "intermediate" distance resulted in an appetitive reward. Faced with the same 2 × 6 grid used for general touchscreen testing, mice were presented with one of two lit squares, either the 2nd or 5th square in the bottom row. Only one of the squares was the rewarded stimulus (S+) at a given time during the daily 30 min session. Touching the correct side led to the same rewarding conditions as during training (reward tray light, tone, and 7 μL of the reward), whereas incorrect touches (touches to S−) led to no reward and a time out period w/ house light on. After reward collection or a time-out end, ITI (10 s) occurred before a new trial began. The new trial started when the mouse poked their nose in the reward tray. The following day, the other lit square became S+ based on the previous day's final trial. Alternation of S+ continued until the mouse reached 50 trials (regardless of accuracy) within 30 min. Reversals occurred when the S+ was correctly chosen in 7 of 8 consecutive trials, and the opposite side of the screen became S+ within the same 30 min testing session. LDR Train completion occurred when the mouse reached at least 2 reversals in 2 out of 3 consecutive days. Mice then moved onto the LDR Test phase. In LDR Test, there were two difficulty settings: "Large Separation", where the S+ and S− stimuli were the most lateral squares (7th and 12th windows on the bottom row of a 2 × 6 grid) and 'Small Separation' where the S+ and S− stimuli were the center-most squares (9th and 10th windows of the bottom row on the same grid window). The difficulty settings at the start of LDR Test were assigned pseudorandomly to each mouse and were counterbalanced within-group. Similar to LDR Train, during LDR Test reversals occurred once 7/8 correct consecutive S+ response were made; the S+ side was then switched to the opposite side of the screen. Separation difficulties alternated every 2 days, and the sessions ended when the mouse completed 81 trials (regardless of accuracy) or 30 min had passed. The LDR Test phase ended after 20 sessions (10 in each Large and Small separation) were completed. These LDR Train data measures were analyzed: % subject to reach criteria, days to completion, and the Last Day performance measurement (completed trial #, session length, percent correct). These LDR Test discrimination learning (Large separation) and pattern separation (Small separation) data measures were analyzed: percent correct to 1st reversal over all 5 blocks (Last Day of each block), and percent correct, trial number, and session length to 1st reversal in the first block (Block 1) and last block (Block 5). These LDR Test reversal learning/cognitive flexibility data measures were analyzed: number of reversals achieved in Large and Small separations in the Last Day of Block 5 as well as in the Last Day of Blocks 1–5.

2.5.4 | Stimulus–response acquisition (Acq) and extinction (Ext) learning

After the completion of LDR Test at 8.6 weeks post-IRR, all mice received normal chow ad libitum for 2 weeks. At the end of 1 week, mice first underwent elevated plus maze (see Arena-based Testing, below). At 9.25 weeks post-IRR, mice were placed back into the touchscreen operant chamber with a 1 × 3 grid for a reintroduction period (10 min, 1 day) which used the Hab 1 program. That evening and 14 h prior to the next touchscreen session (simple stimulus–response acquisition, or Acq), mice were again placed under food restriction.

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Testing for simple stimulus–response acquisition (Acq) and extinction (Ext) learning in an operant touchscreen paradigm began the next day and used the programs Extinction 1 and 2, respectively (Lafayette Life Sciences, cat. no. 89547). For both Acq and Ext, a mouse was faced with a 1×3 grid with the middle-lit rectangle as the S+. Acq: Acq sessions occurred daily. As the mouse was placed into the chamber, the reward tray was illuminated and a reward was provided. Once the mouse nose poked the tray, S+ appeared in the middle of the touchscreen. Once touched, a reinforcing appetitive reward was delivered and a tone was played. Touches to either of the two lateral blank rectangles resulted in no reward and no tone. The mouse collected the reward and an intertrial interval (ITI) began. Then the reward tray was illuminated again to begin the next trial. Mice were tested daily and sessions ended once mice completed 30 trials by touching the conditioned stimulus 30 times or after 30 min had elapsed. Criteria for Acq of a simple stimulus–response included completion of 30 trials in <15 min in four consecutive days. Although each Acq session was 30 min in length, only the first 15 min were considered for completion criteria to ensure that subsequent Ext learning couldl occur (Bussey et al., 2012; Cotter et al., 2023; Mar et al., 2013; Soler et al., 2021). Once completed, each mouse moved directly to the Ext learning sessions. Ext: Ext sessions occurred daily and began with an initial 5-s interval where the screen was blank. The stimulus (middle-lit rectangle) then appeared for ≤10 s. If touched, the stimulus disappeared, no reward was provided, but the food hopper was illuminated and the tone still played. Touches to the blank parts of the screen resulted in no response from the apparatus and the 10-s timer was unchanged. Regardless of omission (no touch at all to screen), blank touch, or stimulus touch, a 10-s interval of no stimulus presentation occurred between trials, but the hopper remained illuminated, and entries to the hopper turned the light off until the stimulus was touched in the following trial. Daily sessions consisted of 30 trials. Mice completed Ext once ≥24 trials were successfully omitted across 3/4 consecutive days. Measures collected for Hab 1 included beam breaks (total from reward tray, two side bars, and screen beam), for Acq included days to completion, session length, correct touch latency, and reward collection latency, and for Ext included days to completion, session length, and response number. Acq performance over time was analyzed on Day 1, Day 4, and Last Day; Day 4 was selected since the Acq criteria requires completion of 30 trials in <15 min on four consecutive days. Ext performance over time was analyzed on Day 1, Day 8, and Last Day; Day 8 was selected since the average group Ext completion was 8 days.

Arena-based testing

For at least 2 weeks before all arena-based tests, mice had ad libitum access to food and water.

2.5.5 | Elevated-plus maze (EPM)

Behavior in the EPM (Harvard Apparatus, cat. no. 760075) was assessed 8.75 mon post-IRR (Figure 1a). Mice were placed in the center of the arena pseudorandomly facing one of two open arms (43×33×6 cm, height×arm length×arm width) and freely allowed to explore the arena for a single 5-min trial. EPM was run under 25 lux (red light) and from 12:55 to 18:33. In Ethovision, arena areas were defined as open arms, closed arms, and the center zone between the four arms, and the center point of each animal was used for tracking. Measures collected for EPM included duration in open and closed arms and distance traveled in open and closed arms. From these measures, an exploration index was calculated:

$$\text{EPM Exploration Index} = \frac{(\text{Closed} - \text{Open Arm})}{(\text{Closed} + \text{Open Arm})}$$

2.5.6 | Gross locomotor activity in a novel environment

Gross locomotor activity was examined 11.5 mon post-IRR (Figure 1a) after Acq and Ext. This test differed from other assessments of locomotion in this experimental timeline in that the mice were placed in a novel sound-isolated chamber for one time. The dimensions of the activity monitor chambers used (MED Associates, cat. no. ENV-510) were 27×27×20 cm and 16 beams detected movement. Mice underwent a single 30-min session in ca. 32 lux (white light) from 09:00 to 16:00. Measures collected by an array of infrared sensors (Activity Monitor 7 software, MED Associates, RRID:SCR_014296) include total distance moved, mean velocity, ambulatory time, ambulatory episodes, rearing time, and number of rearings.

2.5.7 | 3-Chamber social interaction test (3-CSI)

The 3-CSI test was run 11.75 mon post-IRR (Figure 1a) as a one-day, three-trial test where mice are placed in a bespoke three-chamber arena (Kiffer et al., 2021) to assess exploration during habituation, sociability, and social novelty (Crawley, 2007; Shoji & Miyakawa, 2019). Habituation trial: Mice were placed in the empty, novel 3-CSI testing arena to assess inherent bias in the novel testing environment which might entice a mouse to spend more time in the left, middle, or right chamber. There were no objects/enclosures inside the 3-CSI chamber during habituation. *Sociability trial*: A novel conspecific mouse ("Stranger 1") inside a small, perforated enclosure was pseudorandomly placed in the left or right chamber. In the opposite chamber, an empty perforated enclosure ("Empty") is placed. *Social novelty trial*: The enclosure with Stranger 1 (a mouse which is now familiar) was placed back in the same left or right chamber and a second age- and sex-matched stranger conspecific ("Stranger 2") was placed in a perforated enclosure in the opposite chamber. Each of the three trials (habituation, sociability, social novelty) were 10 min each and separated by 1 min. Experimenter gloves were switched frequently throughout 3-CSI and the arena and enclosures were cleaned with MB-10 and allowed to dry in-between test mice. Six age-matched C57BL/6J female stranger mice were used as the social stimuli pseudorandomly such that they were never used in two consecutive

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tests to minimize stress. All trials of 3-CSI were run under 25 lux (red light) and performed from 09:30 to 20:00. The cage tested first was chosen randomly each day, and cages alternated groups throughout the rest of that day. Ethovision arena dimensions and exploration areas are designated as previously described (Kiffer et al., 2021). For analysis of all three trials, arena boundaries were drawn for each chamber (left, middle, right). For analysis of social novelty, a "sniff zone" was designated in the left and right chamber and defined as a 1.5 cm radius around the enclosure (which was either empty or held a stranger mouse). Measures collected for habituation include time spent in the left and right chamber. Measures collected for sociability include time spent in the social stimulus-containing chamber (Stranger 1) and the empty enclosure chamber (Empty). Measures collected for social novelty include time in the sniff zone around the familiar stimulus (Stranger 1) and around the novel social stimulus (Stranger 2).

2.5.8 | Open field (OF) in novel environment

OF was run between 12 and 13.25 mon post-IRR in a custom opaque white polycarbonate arena (Figure 1a; Kiffer et al., 2021). Two days of OF were performed (Day 1, Day 2) on subsequent days to capture exploration and locomotion in a novel environment (Day 1) and in a now-familiar environment (Day 2). Mice were placed in the center of a novel OF arena and allowed free exploration for one 10-min session for Day 1 and Day 2. OF was run under 25 lux (red light). OF Day 1 was run from 11:00 to 17:30, and OF Day 2 was run from 14:00 to 18:30. Exploration zones defined in Ethovision include the four 5 × 5 cm corners (Corner) and the 20 × 20 cm arena center (Center). OF measures collected include distance traveled, time in Center, and Exploration Index:

$$\text{OF Exploration Index} = \frac{(\text{Center} - \text{Corner})}{(\text{Center} + \text{Corner})}$$

Day 1 and Day 2 metrics are shown and analyzed separately as they reflect different measures. As a measure of habituation learning, data are also presented comparing the distance traveled, time in Center, and Exploration Index of Day 1 versus Day 2.

2.5.9 | Novel object recognition (NOR)

The NOR test was performed immediately after Day 2 of OF between 12 and 13.25 mon post-IRR (Figure 1a). NOR is a two-day test, but as it is run in the same arena as OF. Therefore, OF Day 1 and Day 2 served as two habituation trials for NOR, and the two NOR days can be considered Day 3 (familiarization with objects) and Day 4 (test day). On Day 3, two identical objects were placed in the center of the OF arena and mice freely explored for 10 min. The two identical objects were 50 mL blue-capped centrifuge tubes filled with blue nitrile gloves and water. On Day 4, one of the two objects was replaced with a novel object, a 200 mL polycarbonate cell culture flask filled with light white aquarium pebbles, and the mouse explored for another 10 min. NOR was run under 25 lux (red light) and from 13:30 to 19:00. The arena, object dimensions, and exploration boundaries used for analysis are as previously described (Kiffer et al., 2021). Objects used were deliberately heavy to prevent mice from climbing on or pushing the objects and were made of common laboratory equipment to enable reproducibility. Measures collected for NOR include object exploration time on Day 4 (Familiar and Novel). From these data, object discrimination index was calculated as follows:

$$\text{NOR Discrimination Index} = \frac{(\text{Novel} - \text{Familiar})}{(\text{Novel} + \text{Familiar})}$$

2.5.10 | Marble burying in a home cage

Marble burying activity was measured between 12 and 13 mon post-IRR (Figure 1a) as a measure of repetitive, compulsive-like behavior independent of anxiety or exploratory behaviors (Luigjes et al., 2019; Thomas et al., 2009). We used standard glass marbles (Ø = 15 mm) placed on 3 cm of standard rodent hardwood bedding (Beta Chip, Northeastern Products Corp.) in a 20 × 35 × 15 cm arena. Mice were allowed to freely interact with marbles in a single 20-min trial. At the end of the trial, mice were returned to their home cages. Marble burying was run in ca. 25 lux (red light) and from 13:30 to 17:00. Measures collected for marble burying include the number of marbles buried which was evaluated by the experimenter. Marbles that were approximately ≤25% visible were considered buried.

2.5.11 | Nestlet shredding in a home cage

Nestlet shredding was measured between 12 and 13 mon post-IRR (Figure 1a) as a complementary method of measuring compulsive-like behaviors (Jeyabalan et al., 2022). At 20:00, mice were singly housed into identical cages with standard chow and water ad libitum. Mice were given pre-weighed nestlets (Animal Specialties and Provisions, ANNES) and no additional enrichment. Cages remained on the rack in the housing room overnight. Twelve hours later, mice were placed back into their homecage. Measures collected for nestlet shredding include the weight of the intact portions of the nestlets, with intact portions operationally defined by the experimenter.

2.6 | Tissue collection and processing

Approximately 1 mon after the completion of behavioral tests (14.25 mon post-IRR), cages of mice were brought one at a time into the procedure room where mice were decapitated within 1 min of being brought into the room. Brains were promptly removed from the skull and immersion fixed in

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4% paraformaldehyde (Sigma-Aldrich, cat. no. 441244) made in 1xPBS (Phosphate Buffered Saline, 10X Solution, Fisher Scientific, cat. no. BP39920). The following day, paraformaldehyde solution was replaced with fresh solution, and the day after brains were immersed in 1xPBS with 30% sucrose (Fisher Scientific, cat. no. S5-3) and 0.01% NaN_3 (Sigma-Aldrich, cat. no. S8032) and stored at 4°C until sectioning. Brains were sectioned (40 μm) in the coronal plane on a freezing microtome in preparation for stereological assessment of cells throughout the hippocampus, as previously reported (Lagace et al., 2010). Sections were stored at 4°C in 1xPBS with 0.01% NaN_3. Prior to staining, sections of the hippocampus sampled at every 360 μm (1:9 series) were mounted on superfrost-plus slides and dried at room temperature (RT) overnight.

2.7 | Immunohistochemistry (IHC)

A random selection of mice that went through behavior were selected for doublecortin (DCX) IHC ($n = 11$/group). Slide-mounted IHC was performed as previously described (Yun et al., 2023). In brief, slide-mounted sections were placed into near-boiling citric acid (pH 6.0; Fisher Scientific, cat. no. A940) for antigen unmasking. Endogenous peroxidases were inhibited via incubation in 0.3% hydrogen peroxide (Sigma-Aldrich, RRID:CHEBI_16240) at 30 min RT. Blocking of nonspecific binding sites was accomplished by incubation in blocking solution (3% normal donkey serum, Jackson ImmunoResearch, RRID:AB_2340396; 0.3% Triton X-100, VWR, RRID:CHEBI_9750) in 1XPBS for 1 h at RT. Sections were incubated overnight at RT in goat-α-doublecortin (1:8000, Santa Cruz Biotechnology, RRID:AB_2088494) in 3% normal donkey serum and 0.3% Tween 20 (VWR, cat. no. 200002-836). After 1xPBS washes, sections were incubated in biotinylated-donkey-α-goat-IgG (1:200, Jackson ImmunoResearch, RRID:AB_2340396) for 1 h at RT. After additional 1xPBS washes, the signal was amplified and HRP-tagged via incubation in avidin-biotin complex (Vectastain Elite ABC-HRP Kit, Vector Laboratories, cat. no. PK-6100) for 1 h at RT. The HRP signal was colorimetrically visualized via timed incubation in metal-enhanced diaminobenzidine (Thermo Fisher Scientific, cat. no. 34065) at RT. Counterstaining was achieved via Nuclear Fast Red Counterstain (Vector Laboratories, cat. no. H-3403). Sections were dehydrated by incubation in a series of increasingly concentrated ethanol washes followed by delipidization in Citrisolv (Fisher Scientific, cat. no. 04-355-121). Coverslipping was done with DPX (Electron Microscopy Services, cat. no.13512) and 24 × 60 mm coverglass (VWR, cat. no. 48393-251).

2.8 | Stereological cell counts

DCX-immunoreactive (DCX+) cells were quantified in one hemisphere of the subgranular zone (SGZ) of the dentate gyrus granule cell layer (GCL). An observer blind to experimental conditions counted all SGZ DCX+ cells by using an Olympus® BX-51 brightfield microscope (cat. no. BX-51) with 400x magnification (40x objective, N/A) and examining the entire focal (Z) plane. Stereological principles were followed (Lagace et al., 2010; Whoolery et al., 2017). Quantification was performed in the SGZ along the entire anterior–posterior hippocampal axis (−0.82 to −4.33 mm from bregma). When counted, cells were categorized as DCX+ immature neurons (brown-stained soma located in SGZ, presence of neurite and dendrite outgrowth, some dendritic branching) or DCX+ progenitor cells (brown-stained soma located in the SGZ, no neurite outgrowth). For total cell number, the value collected for each mouse was multiplied by the section sampling fraction (here 1/18 since one hemisphere was counted). For cell numbers across bregma, the raw value collected at each bregma level is provided.

2.9 | Computer scripts

A custom Python (version 3.8.6, RRID:SCR_008394) script developed by the Eisch Lab was used to extract touchscreen data, calculate outcome values, compile the data into a database, and export filtered data into relevant csv files for analyses (Yun et al., 2023). These scripts and sample data files are available at https://github.com/EischLab/ts-data-analysis-app. Extracting the data into an output CSV file was managed with another custom script, and these outputs were verified manually. Following this verification, the data were analyzed using GraphPad Prism 10 (RRID:SCR_002798) according to the tests detailed in the Statistical Analysis section. Survival analyses were performed in R, and a full analysis notebook is found at https://github.com/EischLab/19AFemalesSurvival.

2.10 | Randomization, blinding, subject number, and data removal

To ensure equal sample size in each of our four predetermined group (Veh/Sham, Veh/33-GCR, CDDO-EA/Sham, CDDO-EA/33-GCR), the 64 purchased mice were divided into groups based on "block randomization" (Efird, 2011; Kang et al., 2008; McEntegart, 2014); here this is referred to as "group randomization" to avoid confusion with the term Block (which is used in the LDR behavioral task). Each cage of 4 mice was arbitrarily assigned to one of the four groups prior to the start of the experiment, resulting in 16 mice (4 cages) per group. After group assignment, cages of mice were weighed to ensure similar mean body weight per group. For examination of DCX+ cells, subjects ($n = 11$/group) were arbitrarily chosen among mice that underwent behavior tests for DCX+ cell number quantification. Experimenters were not blinded to the variable Diet or Radiation at BNL, but were blinded to these variables throughout the remainder

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of the experiment. The subject number of each group in each experiment is provided in Table S1. The following rules applied for data removal/censorship. Mice were considered for removal from the entire study if they had husbandry issues ($n=2$, Veh/Sham mice, flooded cage on week 6) or veterinarian recommendation ($n=3$; $n=1$ CDDO-EA/33-GCR mouse, hindlimb wounds on week 5; $n=1$ CDDO-EA/Sham mouse, severe dermatitis and hunched posture on week 51; $n=1$ CDDO-EA/33-GCR mouse, underweight, abnormal gait, and hunched posture on week 55). Ultimately, data from the 3 mice that were considered for removal early in the study ($n=2$ Veh/Sham mice from week 6 and $n=1$ CDDO-EA/33-GCR mouse from week 5) were excluded from the entire study. Data from the remaining two mice were included in all graphs up until their removal from the study in weeks 51 and 55, respectively. For general touchscreen training and LDR studies, the subject numbers were: Veh/Sham ($n=14$), Veh/33-GCR ($n=16$), CDDO-EA/Sham ($n=16$), CDDO-EA/33-GCR ($n=15$). For EPM, the subject numbers were: Veh/Sham ($n=14$), Veh/33-GCR ($n=16$), CDDO-EA/Sham ($n=16$), CDDO-EA/33-GCR ($n=14$; one CDDO-EA/33-GCR mouse stepped off of the EPM). For Acq, the subject numbers were: Veh/Sham ($n=12$; two did not meet Acq criteria), Veh/33-GCR ($n=13$; three did not meet Acq criteria), CDDO-EA/Sham ($n=14$; two did not meet Acq criteria), CDDO-EA/33-GCR ($n=14$; two did not meet Acq criteria). For Ext, the subject numbers were: Veh/Sham ($n=12$; carried through from Acq), Veh/33-GCR ($n=12$; 13 carried through from Acq, and one additional did not meet Ext criteria), CDDO-EA/Sham ($n=14$; carried through from Acq), CDDO-EA/33-GCR ($n=14$; carried through from Acq). For measures collected in general activity chambers, there were 6 subjects whose data were censored: 5 because of equipment failure (4 Veh/33-GCR, 1 Veh/Sham) and 1 CDDO-EA/Sham mouse because of outlier status. For 3-CSI, in the social novelty phase, 1 Veh/Sham mouse was excluded because of complete lack of movement, and 2 Veh/33-GCR mice were excluded because of equipment failure. For 3-CSI, the data from one Veh/Sham mouse were not collected during habituation because of equipment failure, and therefore the mouse's entire data set (including sociability and social novelty) was removed from 3-CSI. During 3-CSI social novelty, the data from an additional Veh/Sham and two Veh/33-GCR mice were censored because of equipment failure. For OF, the datate from four Sham/CDDO-EA and four Veh/33-GCR mice were censored because of equipment failure on Days 1 and 2, respectively. Also, the data from one additional Sham/CDDO-EA mouse and one Veh/Sham subject were removed because of results of robust outlier testing (ROUT).

2.11 | Statistical analysis

ROUT (Q = 1%) was performed for all measures. For normality testing, within-group distributions were assessed by the D'Agostino-Pearson test and/or Shapiro–Wilk test and visualized via Quantile-Quantile plots. Most data were normally distributed and thus parametric statistics were used for main effects and interactions followed by multiple comparison Tukey or Bonferroni post-hoc analysis depending on the balance between statistical power and type I error rate (Kiffer et al., 2021; Lee & Lee, 2020; Schaeffer et al., 2022). For interactions, all results are reported and none are excluded. If data were not normally distributed, non-parametric testing was done via Kruskal–Wallis tests for main effects of Treatment (combined variable of Diet/IRR) followed by Dunn's Correction for post-hoc analysis. If samples were missing or data was censored, a mixed model ANOVA was used. Additional non-parametric testing was done for survival and touchscreen completion measures, which were evaluated by the Log-Rank (Mantel-Cox) test. Analyses were performed on the following variables (levels provided in parentheses): Diet (Veh, CDDO-EA), IRR (Sham, 33-GCR), Bregma (12 distances from bregma). Test-specific statistical variables with comparisons included: for 3-CSI, Exploration time (Left vs. Right Chamber), Sniff Zone (Stranger 1 vs. Empty, Stranger 1 vs. Stranger 2); for NOR, Object (Familiar vs. Novel). LDR Test analysis used a combined variable of Treatment (Diet/IRR) and repeated measures (Block), with subsequent Block analysis performed using a two-way ANOVA of Diet×IRR. Other analyses that used a combined variable of Treatment (Diet/IRR) included stimulus–response Acq and Ext, bregma analysis for DCX+ cells and alopecia analysis. Full survival and weights statistical analysis: https://github.com/EischLab/NSRL19AFemales. For effect size determination after ANOVA, partial omega squared (ω^2) or generalized eta squared (ges; η^2) are reported with ≤0.01 small effect size, ≤0.06 medium effect size, and ≤0.13 large effect size (Lakens, 2013; Olejnik & Algina, 2003) using the doomlab applet (https://doomlab.shinyapps.io/mote/) or RStudio (RRID:SCR_001905); 95% confidence interval (CI) is reported up to two significant digits. For effect size determination after Kruskal–Wallis, partial eta squared is reported with ≤0.01 small effect size, ≤0.06 medium effect size, and ≤0.14 large effect size (Lakens, 2013; Olejnik & Algina, 2003). If the 95% CI non-abbreviated value includes zero, the effect size is considered similar to zero (Table S1), although post-hoc analyses are still performed if the main effect or interaction of p value(s) is/are <0.05. For effect size determination of post-hoc analysis of multiple comparisons, Cohen's d and Cliff's delta are reported for ANOVA results and Kruskal–Wallis test, respectively (https://www.psychometrica.de/effect_size.html). In the Results section, values for F, p, partial ω^2, partial η^2, post-hoc P, Cohen's d, and Cliff's delta are reported only when ANOVA (main effects and/or interaction) or Kruskal–Wallis test resulted in $p<0.05$. However, all statistics for all data are provided in Table S1. Data in line graphs are expressed as mean+/-SEM, data in violin plots are expressed as median+/−interquartile range, and data in Kaplan–Meier curves are expressed as +/−95% CI for survival. Percent change of a given effect was calculated from means which are provided in Table S1. Critical α was set to 0.05 for all test statistics. Significant p values are bolded in Table S1 and reported unrounded to the third decimal. For main effects and interactions, the following applies: $^{*}p<0.05$, $^{**}p<0.01$, $^{***}p<0.001$, and $^{****}p<0.0001$. For post-hoc and multiple comparisons, a lower case, italicized letter indicates specific group differences: *a*, Veh/Sham vs. Veh/33-GCR; *b*, Veh/Sham vs. CDDO-EA/Sham; *c*, Veh/

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Sham vs. CDDO-EA/33-GCR; *d*, Veh/33-GCR vs. CDDO-EA/Sham; *e*, Veh/33-GCR versus CDDO/33-GCR; *f*, CDDO-EA/Sham vs. CDDO-EA/33-GCR. The *p* value for post-hoc analyses is indicated by the number of apostrophes after the lower case italicized letter: *x* $p<0.05$, *x'* $p<0.01$, *x"* $p<0.001$, and *x"'* $p<0.0001$. Post-hoc values of $0.05<p<0.08$ are only considered notable (and indicated on graph by #) when two additional conditions are met: if effect size is medium or larger and if post-hoc 95% CI does not include zero (Table S1). Otherwise, the $0.05<p<0.08$ is listed in Table S1 but not considered relevant for data interpretation. Statistics were computed using Prism 10 (Graphpad, RRID:SCR_002798) and R studio. A complete report of distributions, test statistics for main effects, *p* values, intergroup comparisons, corrections, sample sizes and effect sizes per measure are provided in Table S1.

2.12 | Graphs, schematics and photomicrographs

Graphs were generated in Prism 10 and RStudio. Brightfield photomicrographs were collected with an Olympus digital camera and cellSens Standard software (RRID:SCR_014551). The calibrated scale bar was photographed at the same objective used for the photomicrographs. Original photomicrographs were cropped in Adobe Photoshop (RRID:SCR_014199), and the crop was pasted into a new Photoshop window with a 300-dpi resolution and saved as a .tif. The .tif was then placed into the figure layout in Adobe Illustrator and scaled down to final size. Final figures and schematics were generated in Adobe Illustrator (v2023, RRID:SCO_010279) or Biorender (RRID:SCR_018361).

3 | RESULTS

The entire experimental period from mouse arrival at BNL at 6-mon-of-age to tissue collection at 20.25-mon of age (14.25 mon post-IRR) was 8.25 mon (Figure 1a, Table S1). At no time was the weight of Veh/33-GCR mice different from the weight of Veh/Sham mice (Figure 1b,c; Table S1). Beginning two-month post-IRR, all mice received food during a restricted period of time in preparation for operant touchscreen training and testing. At one time point before and most time points after food restriction, CDDO-EA/Sham mice showed lower weight relative to Veh/Sham mice, and at two time points CDDO-EA/33-GCR mice weighed on average 9% less than CDDO-EA/Sham mice (Figure 1b, Table S1; interactions: IRR×Time $F(4.35, 243.64)=2.692$, $p=0.028$ and Diet×Time $F(4.35243.64)=3.710$, $p<0.001$; post-hoc: Veh/Sham versus CDDO-EA/Sham: *b* $p<0.05$, *b'* $p<0.01$ and CDDO-EA/Sham vs. CDDO-EA/33-GCR: *f* $p<0.05$). For the weight collected at the end of the experiment (14.25 mon post-IRR), CDDO-EA/Sham mice weighed on average 12.5% less than Veh/Sham mice (Figure 1c, Table S1; main effect: Diet $F(1, 55)=5.057$, $p=0.028$, partial $\omega^2=0.06$ (Medium), 95% CI [0.00, 0.23]; post-hoc: Veh/Sham vs. CDDO-EA/Sham: *b#* $p=0.0652$, Cohen's $d=0.885$ (Large), 95% CI [−1.43, −0.111]). These body weight data show three things: (1) 33-GCR exposure at 6-mon-of-age on its own does not change body weight of female mice when assessed over the following 14.25 mon; (2) CDDO-EA given for 5 days at 6-mon-of-age slightly decreases body weight for much of the experiment between 2 and 14.25 mon post-IRR; and (3) exposure to 33-GCR eliminates this CDDO-EA-induced lower weight.

The four groups did not differ in overall survival (Figure S1; Table S1). The groups did differ in coat score collected 10.5 mon (42 weeks) and 13.5 mon (54 weeks) post-IRR. At 42 weeks, CDDO/33-GCR mice had worse alopecia than Veh/Sham mice and at 54 weeks, Veh/33-GCR and CDDO-EA/Sham mice had worse alopecia than Veh/Sham mice (Figure S1, Table S1; Treatment $H=40.55$, $p<0.0001$, partial $\omega^2=0.23$ (Large); post-hoc 42 weeks: Veh/Sham vs. CDDO-EA/33-GCR, *c* $p=0.010$, Cliff's delta = 0.8 (Large), 95% CI [0.467, 0.933], 54 weeks: Veh/Sham vs. Veh/33-GCR: *a'* $p=0.007$, Cliff's delta = 0.728 (Large), 95% CI [0.357, 0.911], 54 weeks: Veh/Sham vs. CDDO-EA/Sham: *b* $p=0.0234$, Cliff's delta = 0.714 (Large), 95% CI [0.324, 0.895]). These data suggest the exposure to 33-GCR or CDDO-EA diet at 6 mon-old may contribute worse coat condition near the end of the experiment (13.5 mon-old or 54 week-old mice).

Locomotion in the touchscreen chambers during Hab 1 was the same among all four groups (Figure 2a), indicating no pre-existing differences in movement in the novel touchscreen chambers. There were six general touchscreen phases, and CDDO-EA influenced the number of days it took to reach the criterion in two phases: MT and PI (Figure 2b). In MT, CDDO-EA/Sham mice completed MT on average 2.8 days faster than Veh/Sham mice (Figure 2b, Table S1; main effect: Diet $F(1, 57)=13.15$, $p=0.0006$, partial $\omega^2=0.17$ (Large), 95% CI [0.03, 0.36]; post-hoc: Veh/Sham vs. CDDO-EA/Sham: *b* $p=0.015$, Cohen's $d=0.851$ (Large), 95% CI [−1.43, −0.119]). Also, CDDO/33-GCR mice completed MT on average 2.4 days faster than Veh/33-GCR mice (Figure 2b; Table S1; post-hoc: Veh/33-GCR vs. CDDO/33-GCR: *e* $p=0.041$, Cohen's $d=1.09$ (Large), 95% CI [−1.72, −0.372]). In PI, there was a main effect of the CDDO-EA diet, but no post-hoc differences (Figure 2b; Table S1; main effect: Diet $F(1, 57)=6.838$, $p=0.011$, partial $\omega^2=0.09$ (Medium), 95% CI [0.00, 0.26]). PI is distinct from other general touchscreen training phases in that accuracy—in addition to days to completion—can be measured. During PI, CDDO-EA influenced measures relevant to accuracy. Specifically, CDDO-EA/33-GCR mice took on average 44.4% fewer trials to reach criteria relative to Veh/33-GCR (Figure 2c; Table S1; main effect: Diet $F(1, 57)=10.79$, $p=0.001$, partial $\omega^2=0.14$ (Large), 95% CI [0.02, 0.32]; post-hoc: Veh/33-GCR vs. CDDO-EA/33-GCR: *e* $p=0.011$, Cohen's $d=0.845$ (Large), 95%CI [−1.27, −0.273]) and made on average 50.9% fewer blank touches relative to Veh/33-GCR (Figure 2d; Table S1; main effect: Diet $F(1, 57)=10.98$, $p=0.001$, partial $\omega^2=0.14$ (Large), 95% CI [0.02, 0.33]; post-hoc: Veh/33-GCR vs. CDDO-EA/33-GCR: *e'* $p=0.008$, Cohen's $d=0.885$ (Large), 95% CI [−1.32, −0.244]). Also in PI, CDDO-EA/Sham mice made correct touches on average 22.7% faster (had a shorter latency) relative to Veh/Sham mice (Figure 2e; Table S1; main effect: Diet $F(1, 57)=11.71$, $p=0.001$, partial $\omega^2=0.15$ (Large),

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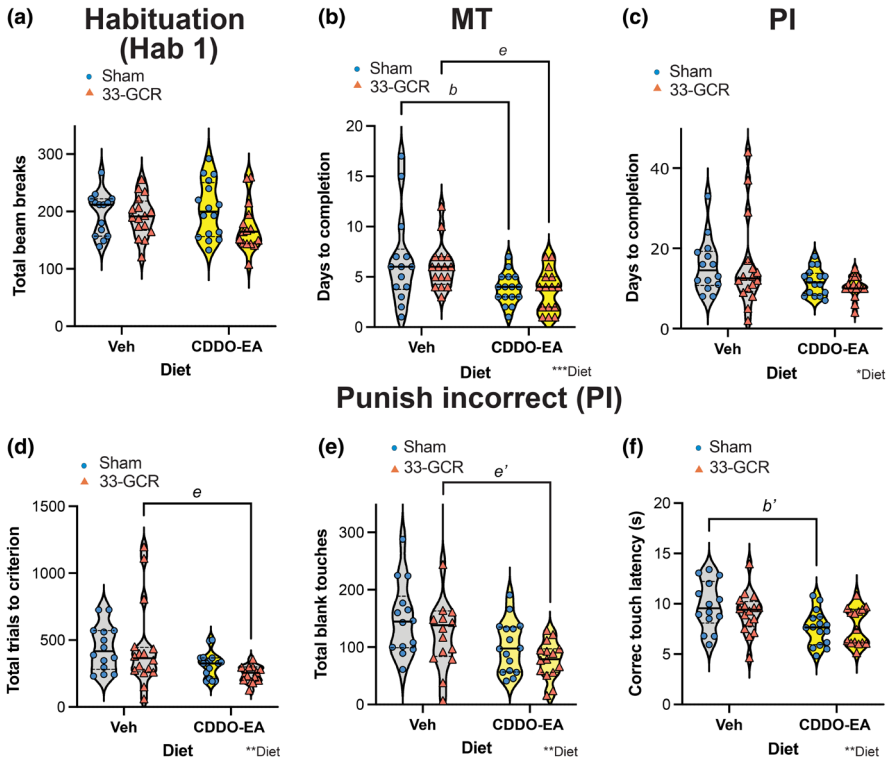


FIGURE 2 Two and one-half mon post-IRR, total beam breaks in a novel touchscreen operant chamber were similar among all groups, but some measures of general touchscreen performance were decreased by CDDO-EA, suggesting improved/faster learning. (a) Beam breaks in novel environment. This is the first time the mice were ever in the touchscreen chamber (the 10-min Habituation 1 [Hab1]). (b, c) Days to Completion for general touchscreen training phases must touch (MT) and punish incorrect (PI). (d–f) Measures relevant to accuracy from PI. Trials to reach criteria (d), Blank touches (e). Correct touches (f). Statistical analysis in a–f: Two-way ANOVA (Diet × IRR). Main effect and interaction are denoted by $^{*}p<0.05$, $^{**}p<0.01$, $^{***}p<0.001$, post-hoc multiple comparison analysis significance denotations between groups: Veh/Sham versus CDDO-EA/Sham, b $p<0.05$ and b' $p<0.01$; Veh/33-GCR versus CDDO-EA/33-GCR, e $p<0.05$ and e' $p<0.01$. 14–16 mice per group. IRR, irradiation. a–e: Veh, vehicle. Violin plots depict median (solid line) and 25% and 75% quartiles (dotted line). Complete subject numbers and detailed data analysis are provided in Table S1.

95% CI [0.02, 0.34]; post-hoc: Veh/Sham vs. CDDO-EA/Sham: b' $p=0.008$, Cohen's $d=1.07$ (Large), 95% CI [−1.88, −0.257]). Together these data suggest that by 2.5 mon post-CDDO-EA and post-IRR, CDDO-EA mice—regardless of whether in the Sham or 33-GCR group—learned the general touchscreen training better/faster than Veh mice.

All mice completed LDR Train (earliest start: 3 mon post-IRR, Figure 3a). All groups did so at a similar rate (Figure 3b; Table S1) and in a similar number of days (Figure 3c; Table S1; interaction: IRR × Diet $F(1, 56)=4.286$, $p=0.043$, partial $\omega^2=0.05$ (Small), 95% CI [0.00, 0.21]; post-hoc: all comparisons $p>0.05$). On the day each mouse completed LDR Train (Last Day), CDDO-EA/33-GCR mice took on average 6.8 (16.2%) more trials compared to Veh/33-GCR mice to reach the criteria and complete LDR Train (Figure 3d; Table S1; main effect: Diet $F(1, 57)=11.90$, $p=0.001$, partial $\omega^2=0.15$ (Large), 95% CI [0.02, 0.34]; post-hoc: Veh/33-GCR vs. CDDO-EA/33-GCR: e' $p=0.003$, Cohen's $d=1.14$ (Large), 95% CI [0.478, 1.88]). However, the remaining LDR Train Last Day metrics—session length (Figure 3e; Table S1; main effect: Diet $F(1, 57)=7.293$, $p=0.009$, partial $\omega^2=0.09$ (Medium), 95% CI [0.00, 0.27]; post-hoc: all comparisons $p>0.05$) and percent correct, a measure of accuracy (Figure 3f; Table S1)—were similar among the four groups. These data suggest that when LDR Train begins (3 mon post-CDDO-EA and post-IRR, 9-mon-old mice), all four groups of mice have roughly similar LDR Train performance metrics, including accuracy.

A major benefit of the LDR Test is that it can have varying "load"; placing the lit squares far apart (Large separation, Figure 4a,b) is low load and considered discrimination while placing the lit squares close together (Small separation, Figure 4c,d) is high load and considered pattern separation (Clelland et al., 2009; Oomen et al., 2013; Soler et al., 2021; Vivar & Praag, 2013; Zhuo et al., 2016). This assessment of discrimination (low load, Large separation) and pattern separation (high load, Small separation) can be measured by the percentage of trials which are correct prior to the first reversal. A second benefit of the five-month-long LDR Test is performance can be assessed over time in "blocks"; each of the five blocks consists of 4 days of LDR Test (2 days Large separation + 2 days Small separation).

On the Last Day of each LDR Test block, mouse accuracy was gauged by percent correct up to first reversal (Figure 4b,d). In Large separation of LDR Test in Block 5, CDDO-EA/33-GCR mice were 26.1% more accurate than Veh/33-GCR mice (Figure 4b; Table S1; Treatment $F(3, 57)=3.079$, $p=0.034$, partial $\omega^2=0.17$ (Large), 95%

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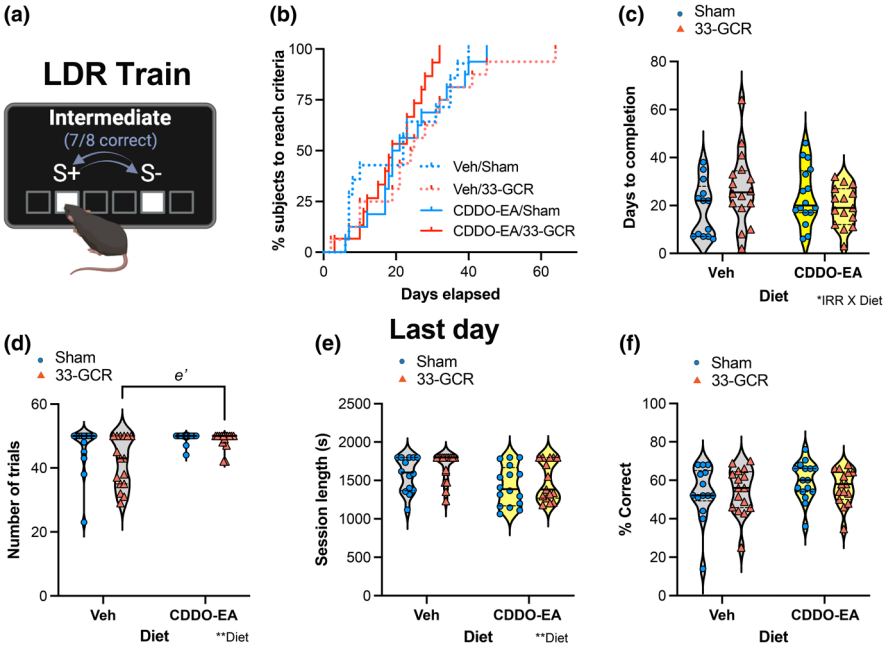


FIGURE 3 Three months post-IRR, performance in location discrimination reversal training (LDR Train) was generally similar among all groups. (a) LDR Train touchscreen set-up. Schematic of the lit squares and their "Intermediate" separation which were used for LDR Train. After a subject made 7 out of 8 correct touches, the windows indicated as S+ and S− switched. (b) Percent of subjects reaching LDR Train criteria over time. The percent of each group to reach criteria for LDR Train was similar among groups. (c) Days to completion for LDR Train. For the number of days it took each group to reach LDR Train completion, there was an IRR × Diet interaction but no post-hoc significance. (d–f) LDR Train performance metrics on the day each mouse reached criteria ("Last Day"). Number of trials completed (d). Session length (e). Percent correct (f). Violin plots depict median (solid line) and 25% and 75% quartiles (dotted line). Statistical analysis in (b): Long-rank (Mantel-Cox) test, with a Kaplan–Meier completion graph and (c–f): Two-way ANOVA (Diet × IRR). Main effect and/or interaction are denoted by $^{*}p<0.05$, $^{**}p<0.01$, post-hoc multiple comparison analysis significance denotations between groups: Veh/33-GCR versus CDDO-EA/33-GCR, *e' p* < 0.01. Violin plots depict median (solid line) and 25% and 75% quartiles (dotted line). 14–16 mice per group. IRR, irradiation. Non-rewarding stimulus, S−, a touch to this lit square did not result in a reward. Rewarding stimulus, S+, a touch to this lit square resulted in a reward. Veh, vehicle. Complete subject numbers and detailed data analysis are provided in Table S1.

CI [0.08, 0.25]; post-hoc: Veh/33-GCR vs. CDDO-EA/33-GCR in Block 5: *e p* = 0.048, Cohen's *d* = 0.979 (Large), 95% CI [0.285, 1.51]). In Small separation of LDR Test, CDDO-EA/Sham mice performed 48.5% better relative to Veh/Sham mice in Block 5 (Figure 4d; Table S1; interaction: Block × Treatment $F(12, 228) = 1.837$, $p = 0.043$, partial $\omega^2 = 0.11$ (Medium), 95% CI [0.01, 0.15]; post-hoc: Veh/Sham vs. CDDO-EA/Sham in Block 5: *b p* = 0.021, Cohen's *d* = 1.11 (Large), 95% CI [0.326, 1.83]). In the same Small separation LDR Test, CDDO-EA/33-GCR mice were more accurate than Veh/Sham mice in Block 2 (52.3%) and 5 (41.5%; Figure 4d; Table S1; post-hoc: Veh/Sham vs. CDDO-EA/33-GCR in Block 2: *c' p* = 0.005, Cohen's *d* = 1.36 (Large), 95% CI [0.59, 2.1]; Veh/Sham vs. CDDO-EA/33-GCR in Block 5: *c' p* = 0.008, Cohen's *d* = 1.32 (Large), 95% CI [0.474, 2.06]). Separate examination of accuracy among groups on Block 1 revealed this improved accuracy was not evident in Block 1 of either Large or Small separation (Figure 4e,g; Table S1). However, on Block 5, in Large separation CDDO-EA/33-GCR mice were on average 26.1% more accurate than Veh/33-GCR mice, although the effect size was small (Figure 4f; Table S1; main effect; Diet $F(1, 57) = 4.283$, $p = 0.043$, partial $\omega^2 = 0.05$ (Small), 95% CI [0.00, 0.20]; post-hoc: Veh/33-GCR vs. CDDO-EA/33-GCR: *e p* = 0.043, Cohen's *d* = 0.979 (Large), 95% CI [0.285, 1.51]). On Block 5 in Small separation, CDDO-EA/Sham mice were on average 48.5% more accurate than Veh/Sham mice (Figure 4h; Table S1; Diet $F(1, 57) = 15.08$, $p = 0.0003$, partial $\omega^2 = 0.19$ (Large), 95% CI [0.04, 0.38]; post-hoc: Veh/Sham vs. CDDO-EA/Sham: *b' p* = 0.004, Cohen's *d* = 1.11 (Large), 95% CI [0.326, 1.83]) and CDDO-EA/33-GCR were on average 30.3% more accurate than Veh/33-GCR mice (Figure 4h; Table S1; post-hoc: Veh/33-GCR vs. CDDO-EA/33-GCR: *e# p* = 0.057, 0.866 (Large), Cohen's *d* = 0.866 (Large), 95%C CI [0.0538, 1.64]). Examination of a complementary variable—session length to first reversal—showed all groups performed similarly in Block 1 for both Large separation (Figure 4i; Table S1; IRR $F(1, 57) = 4.589$, $p = 0.036$, partial $\omega^2 = 0.06$ (Medium), 95% CI [0.00, 0.021], suggesting effect size similar to zero) and Small separation (Figure 4k; Table S1). On the Last Day of Block 5 in Large separation, CDDO-EA/33-GCR mice reached the first reversal on average 54% faster than Veh/33-GCR mice (Figure 4j; Table S1; main effect: Diet $F(1, 57) = 5.201$, $p = 0.026$, partial $\omega^2 = 0.06$ (Medium), 95% CI [0.00, 0.23]; post-hoc: Veh/33-GCR vs. CDDO-EA/33-GCR: *e' p* = 0.008, Cohen's *d* = 1.08 (Large), 95% CI [−1.81, −0.242]). However, on the Last Day of Block 5 in Small separation, CDDO-EA/Sham mice reached their first reversal

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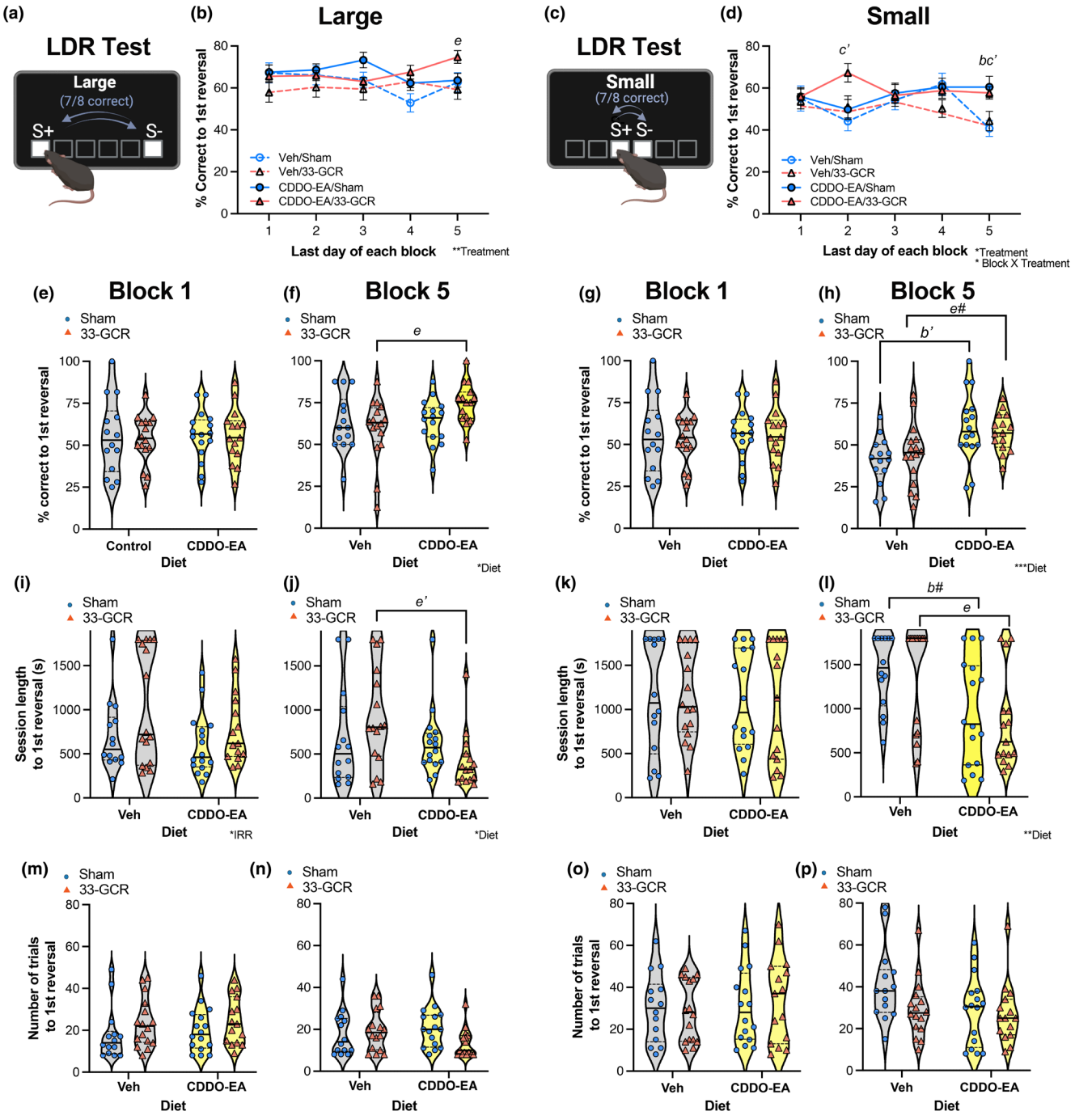


FIGURE 4 Legend on next page

on average 32.3% faster than Veh/Sham mice (Figure 4l; Table S1; main effect: Diet $F(1, 57) = 12.00$, $p = 0.001$, partial $\omega^2 = 0.15$ (Large), 95% CI [0.02, 0.34]; post-hoc: Veh/Sham vs. CDDO-EA/Sham: *b#* $p = 0.052$, Cohen's $d = 0.873$ (Large), 95% CI [−1.64, −0.105]) and CDDO-EA/33-GCR mice reached their first reversal on average 38.1% faster than Veh-33-GCR mice (Figure 4l; Table S1; post-hoc: Veh/33-GCR vs. CDDO-EA/33-GCR: *e* $p = 0.022$, Cohen's $d = 0.906$ (Large), 95% CI [−1.86, −0.0809]. The number of trials to first reversal did not differ among groups on the Last Day of Block 1 or on the Last Day of Block 5 in either Large separation (Figure 4m,n; Table S1) or Small separation (Figure 4o,p; Table S1). These data show that while mice perform LDR Test similarly in Block 1, by Block 5 CDDO-EA/33-GCR mice show better discrimination vs. Veh/33-GCR mice and CDDO-EA mice (regardless of IRR) show better pattern separation vs. Veh mice.

A third benefit of the LDR Test is that the number of "reversals" can be measured, and that reversal number can reflect cognitive flexibility (Swan et al., 2014). The number of reversals are typically analyzed in Blocks at the end of a LDR Test (Soler et al., 2021; Swan et al., 2014; Yun et al., 2023), as mice should be the most familiar

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FIGURE 4 By the end of the 24-day LDR Test, CDDO-EA/33-GCR mice showed better discrimination versus Veh/33-GCR mice, and CDDO-EA mice (regardless of IRR) showed better pattern separation vs. Veh mice. (a, b) Schematic of the lit squares in LDR Test with Large separation (a) and accuracy to 1st reversal over 5 blocks in Large separation (b). Large separation prior to the 1st reversal in LDR Test is considered a test of Discrimination because of the "load low" from the high resolution of the two, maximally separated squares. (c, d) Schematic of the lit squares in LDR Test with Small separation (c) and accuracy to 1st reversal over 5 blocks in Small separation (d). Small separation prior to the 1st reversal in LDR Test is considered a test of Pattern Separation because of the "high load" from the low resolution of the two, minimally separated squares. (e–p) Metrics of Discrimination (Large separation, e, f, i, j, m, n) and Pattern Separation (Small separation, g, h, k, l, o, p) in LDR test were analyzed in Block 1 (e, i, m, g, k, o) and separately in Block 5 (f, j, n, h, l, p) based on accuracy (e–h), session length (i–l), and trial number to the 1st reversal (m–p). Large separation, Block 1, accuracy (e). Large separation, Block 5, accuracy (f). Small separation, Block 1, accuracy (g). Small separation, Block 5, accuracy (h). Large separation, Block 1, session length (i). Large separation, Block 5, session length (j). Small separation, Block 1, session length (k). Small separation, Block 5, session length (l). Large separation, Block 1, trial number to the 1st reversal (m). Large separation, Block 5, trial number to the 1st reversal (n). Small separation, Block 1, trial number to the 1st reversal (o). Small separation, Block 5, trial number to the 1st reversal (p). Statistical analysis in (b, d): Two-way RM measures ANOVA (Block × Treatment), main effect and an interaction followed by Tukey's post-hoc and (e–p): Two-way ANOVA (Diet × IRR), main effect and interaction followed by Bonferroni's post-hoc test. Main effects and/or interaction are denoted by $^{*}p<0.05$, $^{**}p<0.01$, $^{***}p<0.001$. Post-hoc multiple comparison analysis significance denotations between groups: Veh/Sham vs. CDDO-EA/Sham, *b#* $0.05<p<0.08$, *b* $p<0.05$, *b'* $p<0.01$; Veh/Sham vs. CDDO-EA/33-GCR, *c'* $p<0.01$; Veh/33-GCR vs. CDDO-EA/33-GCR, *e#* $0.05<p<0.08$, *e* $p<0.05$, *e'* $p<0.01$. 14–16 mice per group. Non-rewarding stimulus, S−, a touch to this lit square did not result in a reward. Rewarding stimulus, S+, a touch to this lit square resulted in a reward. IRR, irradiation. Veh, vehicle. Violin plots depict median (solid line) and 25% and 75% quartiles (dotted line). Complete subject numbers and detailed data analysis are provided in Table S1.

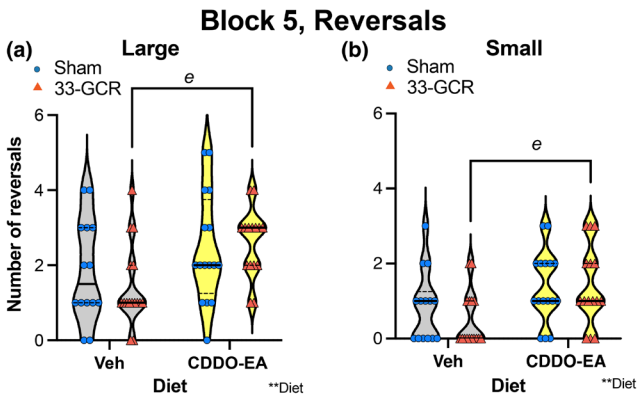


FIGURE 5 By the end of LDR Test (and 3–8.5 mon post-IRR), cognitive flexibility was improved by CDDO-EA during IRR exposure. Number of reversals on the Last Day of the last block (Block 5) of LDR Test under conditions of Large (a) and Small (b) separation. (a) Large separation. (b) Small separation. Two-way ANOVA (IRR × Diet), main effect and/or interaction followed by Bonferroni's post-hoc test. Main effects are denoted by $^{**}p<0.01$. Post-hoc multiple comparison analysis significance denotations between groups: Veh/33-GCR versus CDDO-EA/33-GCR, *e* $p<0.05$. 14–16 mice per group. IRR, irradiation. Veh, vehicle. Violin plots depict median (solid line) and 25% and 75% quartiles (dotted line). Complete subject numbers and detailed data analysis are provided in Table S1.

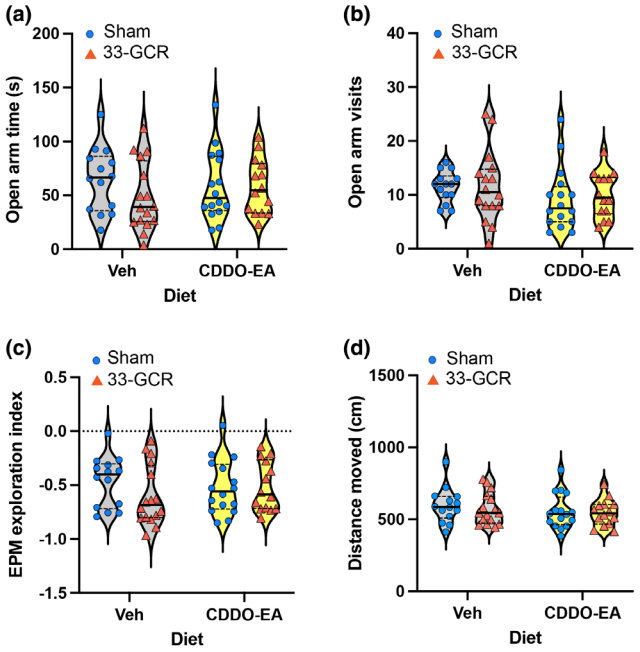


FIGURE 6 Eight months post-IRR, anxiety-like behavior on the elevated plus maze (EPM) was similar among all groups. (a–d) EPM metrics of (a) time in open arms, (b) visits to open arms, (c) exploration index (time spent in open versus closed arms), and (d) distance moved. Two-way ANOVA (IRR × Diet), main effects and interaction are all $p>0.05$ in (a–d). 14–16 mice per group. IRR, irradiation. Veh, vehicle. Violin plots depict median (solid line) and 25% and 75% quartiles (dotted line). Complete data analysis details are provided in Table S1.

with the LDR Test at this point. Thus, our main focus was the number of reversals on the Last Day of Block 5. In Block 5, CDDO-EA/33-GCR mice made on average 80.8% and 180% more reversals than Veh/33-GCR mice during Large separation (Figure 5a) and Small separation (Figure 5b), respectively (Large, Figure 5a; Table S1; main effect: Diet $F(1, 57)=7.725$, $p=0.007$, partial $\omega^2=0.10$ (Medium), 95% CI [0.00, 0.28]; post-hoc: Veh/33-GCR vs. CDDO-EA/33-GCR: *e* $p=0.021$, Cohen's $d=1.15$ (Large), 95% CI [0.28, 2.11]; Small, Figure 5b; Table S1; main effect: Diet $F(1, 57)=8.877$, $p=0.004$, partial $\omega^2=0.11$ (Medium), 95% CI [0.01, 0.30]; post-hoc: Veh/33-GCR vs. CDDO-EA/33-GCR: *e* $p=0.018$, Cohen's $d=0.998$ (Large), 95% CI [0.177, 1.79]). These data suggest that CDDO-EA given at

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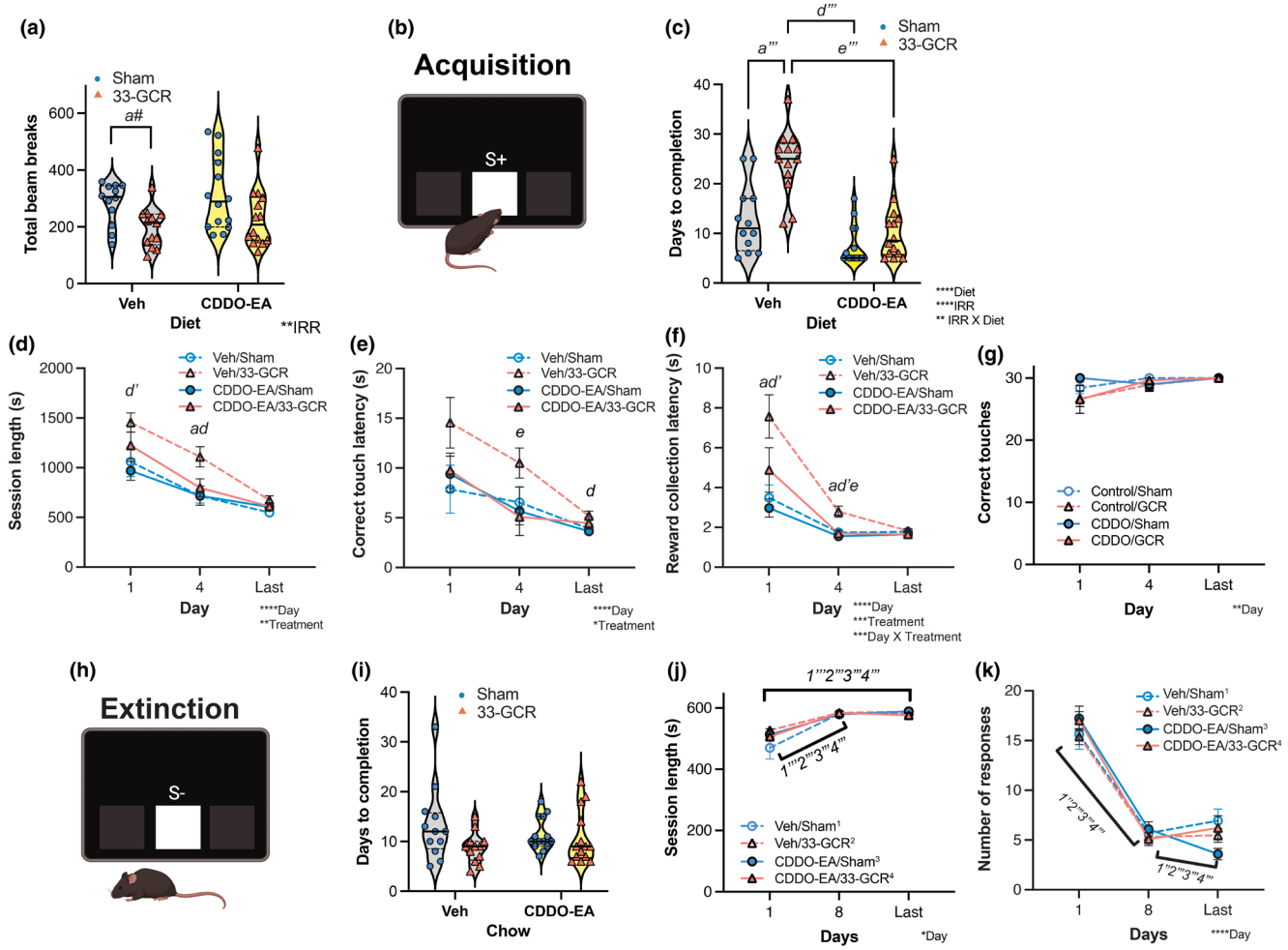


FIGURE 7 9.25 months post-IRR, 33-GCR decreased locomotion in a familiar environment and acquisition of a simple stimulus–response learning task, and CDDO-EA appeared to counter the 33-GCR effects. No effect was seen on subsequent extinction learning. (a) Beam breaks in familiar operant environment. Mice were reintroduced to the touchscreen chamber (10-min Habituation 1 [Hab1]) 3 weeks post-LDR to measure locomotion activity via beam breaks. (b) Schematic of simple stimulus–response learning, "acquisition (Acq)". The middle-lit square (S+) was the conditioned stimulus. Touches to S+ led to reward delivery, while touches to blank areas on either side of S+ did not. (c) Days to complete (acquire) simple stimulus–response learning. Mice are run for 30 min, but only the first 15-min are considered for these data. (d–f) Acq metrics were analyzed in the 1st, 4th, and Last Day based on session length (d), correct touch latency (e), and reward collection latency (f) Note that (d–g) are from the first 30-min of each session. (d) Acq session length. (e) Acq correct touch latency. (f) Acq reward collection latency. (g) Acq correct touches. (h) Schematic of a middle-lit square (S−) in extinction learning. (i–k) Measures of Ext after a simple-response learning based on days to completion (i), session length (j), and response number (#; k). (i) Ext days to completion. (j) Ext session length. (k) Ext number of responses. Statistical analysis: (a, c, i) Two-way ANOVA (Diet × IRR); (d, j, k) Two-way RM (Day × Treatment); (e–g) Mixed Effects 2-way RM ANOVA (Day × Treatment). Main effects and interaction are denoted by $^{*}p<0.05$, $^{**}p<0.01$, $^{***}p<0.001$, or $^{****}p<0.0001$. Bonferroni (a) or Tukey (c–g) post-hoc multiple comparison analysis significance denotations between groups: Veh/Sham vs. Veh/33-GCR, *a#* $0.08<p<0.05$, *a* $p<0.05$, *a'''* $p<0.001$; Veh/33-GCR vs. CDDO-EA/Sham, *d* $p<0.05$, *d'* $p<0.01$, *d'''* $p<0.001$; Veh/33-GCR vs. CDDO-EA/33-GCR, *e* $p<0.05$, *e'''* $p<0.001$; CDDO-EA/Sham vs. CDDO-EA/33-GCR, *f#* $0.08<p<0.05$. Tukey (j, k) post-hoc analysis significance denotations between days within a given group: *1* = within Veh/Sham, *2* = within Veh/33-GCR; *3* = within CDDO-EA/Sham; *4* = within CDDO-EA/33-GCR with *'* = $p<0.001$ and *''* = $p<0.0001$. IRR, irradiation. Veh, vehicle. Violin plots (a, c, i) depict median (solid line) and 25% and 75% quartiles (dotted line). 12–14 mice per group. Complete subject numbers and detailed data analysis are provided in Table S1.

the time of IRR improves cognitive flexibility in 33-GCR mice when tested between 3- and 8-mon post-IRR.

On the EPM, all groups spent similar time in the open arms (Figure 6a; Table S1) and made a similar number of visits to the open arms (Figure 6b; Table S1). All groups had similar EPM exploration indices (time spent in the open vs. closed arms, Figure 6c; Table S1) and moved a similar distance during the 5-min EPM test (Figure 6d; Table S1). These data suggest that at this time point (8.8 mon post-IRR, 14.8-mon-old mice), neither CDDO-EA nor 33-GCR influence this measure of anxiety-like behavior.

Mice were reintroduced to the now-familiar touchscreen chambers for a second 10-min habituation (Hab 1) session as part of the

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training for the subsequent Acq and Ext testing. Veh/33-GCR mice moved on average 30.4% less in these familiar chambers 9.25 mon post-IRR vs. Veh/Sham mice (Figure 7A; Table S1; main effect: IRR $F(1, 49) = 10.01$, $p = 0.002$, partial $\omega^2 = 0.11$ (Medium), 95% CI [0.01, 0.30], post-hoc: Veh/Sham vs. Veh/33-GCR: *a#* $p = 0.063$, Cohen's $d = 1.21$ (Large), 95% CI [−2.22, −0.216]). These data suggest that mice that were exposed to Veh/33-GCR moved less in these familiar chambers 9.25 mon post-IRR compared to Veh/Sham mice, and that this difference is not seen in CDDO-EA/33-GCR vs. CDDO-EA/Sham mice.

After Hab 1, mice began the daily 30-min Acq sessions (Figure 7b). All groups of mice learned to touch the conditioned stimulus within the 30-min sessions (data not shown). However, Acq session data are typically analyzed in only the first 15-min of the session (Bussey et al., 2012; Cotter et al., 2023; Mar et al., 2013; Soler et al., 2021), as the faster learning sets the stage for learning Ext in subsequent sessions. Analysis of the first 15-min of the Acq sessions revealed a visible difference among groups with CDDO-EA/Sham mice learning fastest (mean 7.5 days) and Veh/33-GCR mice taking the longest time to learn (mean 24.4 days, Figure 7c). Further analysis revealed Veh/33-GCR mice took on average 90% more days to complete Acq vs. Veh/Sham (Figure 7c; Table S1; interaction: IRR × Diet $F(1, 49) = 7.457$, $p = 0.0088$, partial $\omega^2 = 0.11$ (Medium), 95% CI [0.00, 0.30], post-hoc: Veh/Sham vs. Veh/33-GCR: *a‴* $p < 0.0001$, Cohen's $d = 1.67$ (Large), 95% CI [0.632, 2.8]). In contrast, two CDDO-EA groups of mice learned Acq faster than Veh/33-GCR mice: CDDO-EA/Sham mice took on average 69% fewer days vs. Veh/33-GCR mice (Figure 7c; Table S1; post-hoc: Veh/33-GCR vs. CDDO-EA/Sham: *d‴* $p < 0.0001$, Cohen's $d = 3.09$ (Large), 95% CI [−4.73, −1.77]) and CDDO-EA/33-GCR mice took on average 57.8% fewer days vs. Veh/33-GCR mice (Figure 7c; Table S1; post-hoc: Veh/33-GCR vs. CDDO-EA/33-GCR: *e‴* $p < 0.0001$, Cohen's $d = 2.28$ (Large), 95% CI [−3.63, −1.12]). Thus, when only the first 15-min were analyzed, as is standard in the field, Veh/33-GCR mice took more days to acquire this simple stimulus–response learning vs. all other groups when examined 9.25 mon post-IRR (15.25-mon-old mice).

To gain greater insight into Acq learning, other metrics of performance from the full 30-min sessions were analyzed on Day 1, Day 4, and Last Day; Day 4 was selected since the Acq criteria requires meeting trial number of four consecutive days. Analysis of the 1st, 4th, and Last Day of each mouse's Acq performance showed three effects on session length (Figure 7d). On Day 1, the session length of Veh/33-GCR mice was on average 33.3% longer than CDDO-EA/Sham mice (Figure 7d; Table S1; main effect of Treatment: $F(3, 49) = 4.892$, $p = 0.004$, partial $\omega^2 = 0.12$ (Medium), 95% CI [0.01, 0.24]; main effect of Day: $F(1.819, 89.15) = 56.32$, $p < 0.0001$, partial $\omega^2 = 0.34$ (Large), 95% CI [0.19, 0.48]; post-hoc: Veh/33-GCR vs. CDDO-EA/Sham on Day 1: *d'* $p = 0.007$, Cohen's $d = 1.37$ (Large), 95% CI [−2.32, −0.45]). On Day 4, Veh/33-GCR mice had session length that was on average 55% longer than Veh/Sham mice (Figure 7d; Table S1; post-hoc: Veh/Sham vs. Veh/33-GCR on Day 4: *a* $p = 0.022$, Cohen's $d = 1.25$ (Large), 95% CI [0.456, 1.98]). Also on Day 4, Veh/33-GCR mice had on average a session length that was 54.5% longer than CDDO-EA/Sham mice (Figure 7d; Table S1; Veh/33-GCR vs. CDDO-EA/Sham on Day 4: *d* $p = 0.046$, Cohen's $d = 1.08$ (Large), 95% CI [−1.97, 0.0495]). In regard to Acq correct touch latency, on Day 4 Veh/33-GCR mice took on average 88.6% longer to make a correct touch compared to CDDO-EA/33-GCR mice (Figure 7e; Table S1; main effect of Day: $F(1.868, 90.62) = 19.64$, $p < 0.0001$; main effect of Treatment: $F(3, 49) = 2.844$, $p = 0.047$; Veh/33-GCR vs. CDDO-EA/33-GCR Day 4: *e* $p = 0.048$, Cohen's $d = 1.11$ (Large), 95% [−1.91, −0.335]), and on the Last Day Veh/33-GCR mice took on average 42.8% longer to make a correct touch compared to CDDO-EA/Sham mice (Figure 7e; Table S1; post hoc Veh/33-GCR vs. CDDO-EA/Sham Last Day: *d* $p = 0.047$, Cohen's $d = 1.11$ (Large), 95% CI [−1.88, −0.299]. In regard to latency to collect the reward during Acq, On Day 1, Veh/33-GCR mice took on average 117% longer to collect the reward compared to Veh/Sham mice (Figure 7f; Table S1; interaction: Day × Treatment: $F(6, 97) = 4.430$, $p = 0.0005$; post hoc: Veh/Sham vs. Veh/33-GCR on Day 1: *a* $p = 0.020$, Cohen's $d = 1.27$ (Large), 95% CI [0.284, 2.02]). Also on Day 1, Veh/33-GCR mice took on average 155% longer to collect the reward compared to CDDO-EA/Sham mice (Figure 7f; Table S1; post hoc: Veh/33-GCR vs. CDDO-EA/Sham Day 1: *d'* $p = 0.006$, Cohen's $d = 1.55$ (Large), 95% CI [−2.24, −0.743]). On Day 4, Veh/33-GCR mice took on average 59.9% longer to collect the reward compared to Veh/Sham mice (Figure 7f; Table S1; post hoc: Veh/Sham vs. Veh/33-GCR Day 4: *a* $p = 0.020$, Cohen's $d = 1.27$ (Large), 95% CI [0.493, 1.9]), Veh/33-GCR mice took on average 79.7% longer to collect the reward compared to CDDO-EA/Sham mice (Figure 7f; Table S1; post hoc: Veh/33-GCR vs. CDDO-EA/Sham Day 4: *d'* $p = 0.004$, Cohen's $d = 1.69$ (Large), 95% CI [−2.34, −0.981]), and Veh/33-GCR mice took on average 67.3% longer to collect the reward compared to CDDO-EA/33-GCR mice (Figure 7f; Table S1; post hoc: Veh/33-GCR vs. CDDO-EA/33-GCR Day 4: *e* $p = 0.010$, Cohen's $d = 1.41$ (Large), 95% CI [−2.1, −0.577]). Mice of all groups showed similar correct touches on Days 1, 4 and the Last Day (Figure 7g; Table S1). These data show three things: (1) 33-GCR exposure slowed acquisition learning of conditioned stimulus learning 9-mon post-IRR; (2) concomitant administration of CDDO-EA at the time of IRR normalized key aspects of this learning; and (3) the negative impact of 33-GCR is on speed of completing the task, and not on accuracy.

Following Acq, mice are presented with similar opportunities to respond to a visual stimulus, but no reward and no cues that are associated with the reward. The time it takes and the extent to which the mice suppress their response allows insight into their ability to extinguish prior learning. Ext performance over time was analyzed on Day 1, Day 8, and Last Day; Day 8 was selected since the average group Ext completion was ca. 8 days. For Ext of the conditioned stimulus learning (Figure 7h), all groups of mice learned to extinguish their simple stimulus–response association and did so over a similar number of days (Figure 7i; Table S1). Analysis of the 1st, 8th, and Last Day of each mouse's Ext performance showed within each group of mice, the length of the sessions on Day 8 and Last Day were longer than the Day 1 session (Figure 7j; Table S1; main effect of Day: $F(1.880, 90.23) = 196.9$, $p < 0.0001$, partial $\omega^2 = 0.72$ (Large), 95% CI [0.60,

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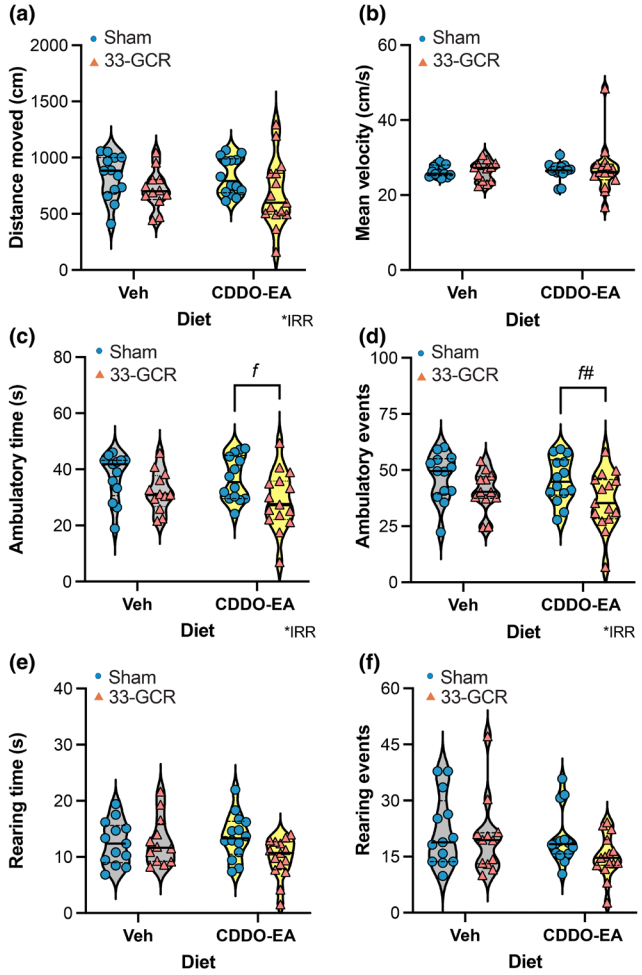


FIGURE 8 Eleven and one-half mon post-IRR, locomotor activity in a novel environment was grossly similar among groups. (a–f) Mice were introduced into a sound-proof square chamber for 30 min and locomotion metrics were measured by broken infrared beams (a) Total distance moved. (b) Mean velocity. (c) Ambulatory time (seconds, s). (d) Ambulatory events. (e) Rearing time (s). (f) Rearing events. (a–f) Two-way ANOVA (Diet × IRR). Main effect is denoted by $^*p < 0.05$. Bonferroni post-hoc multiple comparison analysis significance denotations between groups: CDDO-EA/Sham vs. CDDO-EA/33-GCR, *f* $p < 0.05$, *f#* $0.08 < p < 0.05$. 12–15 mice per group.IRR, irradiation. Veh, vehicle. Violin plots depict median (solid line) and 25% and 75% quartiles (dotted line). Complete data analysis details are provided in Table S1.

0.80]; post hoc information in Table S1) and the number of responses in the sessions on Day 8 and Last Day were less than the Day 1 session (Figure 7k; Table S1; main effect of Day: $F(1.880, 90.23) = 196.9$, $p < 0.0001$, partial $\omega^2 = 0.72$ (Large), 95% CI [0.60, 0.80]; post hoc information in Table S1). Together these data—longer sessions with fewer responses—suggest all mice underwent Ext learning when tested ca. 10- to 11-mon post-IRR (at 16- to 17-mon-of age). As there was no difference in Ext performance across the groups, these data also show no overt effect of 33-GCR or CDDO-EA on Ext.

11.5 mon post-IRR and 1 day prior to 3-CSI, the movement of mice was collected while they spent 30 min in a novel activity chamber (Figure 8). The four groups of mice moved a similar distance (Figure 8a; Table S1; main effect of IRR: $F(1, 50) = 4.737$, $p = 0.0343$, partial $\omega^2 = 0.06$ (Medium), 95% CI [0.00, 0.24]; post hoc: all comparisons $p > 0.05$). All groups were also similar in average velocity (Figure 8b; Table S1). CDDO-EA/33-GCR mice spent on average 16.8% less time ambulating and had 16.8% fewer ambulatory events than CDDO-EA/Sham mice (ambulation time, Figure 8c; Table S1; main effect of IRR: $F(1, 50) = 7.612$, $p = 0.008$, partial $\omega^2 = 0.11$ (Medium), 95% CI [0.00, 0.30]; post hoc CDDO-EA/Sham vs. CDDO-EA/33-GCR: *f* $p = 0.037$, Cohen's $d = 0.839$ (Large), 95% CI [−1.55, −0.0436]; Figure 8d; Table S1; IRR: $F(1, 50) = 6.606$, $p = 0.0132$, partial $\omega^2 = 0.09$ (Medium), 95% CI [0.00, 0.028]; post hoc CDDO-EA/Sham vs. CDDO-EA/33-GCR: *f#* $p = 0.062$, Cohen's $d = 0.774$ (Medium), 95% CI [−1.46, −0.00907]). All four groups spent a similar amount of time rearing (Figure 8e; Table S1) and reared a similar number of times (Figure 8f; Table S1). These data suggest that movement in a novel environment was not grossly impacted by 33-GCR or CDDO-EA 11.5 mon post-IRR (16.5-mon-old mice), although CDDO-EA in the context of 33-GCR did decrease ambulation metrics vs. CDDO-EA in the context of Sham IRR.

During habituation to the novel 3-CSI chamber (Figure 9a), mice in each group showed no preference for either the left or right chamber (Figure 9a; Table S1; main effect of Chamber: $F(1, 55) = 13.60$, $p = 0.0005$, partial $\omega^2 = 0.17$ (Large), 95% CI [0.03, 0.37]; post hoc: all comparisons $p > 0.05$). Although these mice were 17- to 18-mon-old at the time of the 3-CSI sociability trial (Figure 9b), mice in all groups still spent more time in the chamber with the novel mouse (Stranger 1) vs. the chamber with the empty enclosure (Figure 9b; Table S1; main effect of Chamber: $F(1, 55) = 86.05$, $p < 0.0001$, partial $\omega^2 = 0.58$ (Large), 95% CI [0.39, 0.72]; post hoc: all within group comparisons $p \leq 0.005$, all effect sizes Large, all 95%CI exclude 0). This shows all groups of mice, regardless of Diet or IRR, prefer a novel social stimulus more than a novel in animate object, as has previously been shown in older mice (Shoji & Miyakawa, 2019). In the final 3-CSI trial, which tests social novelty (Figure 9c), mice in all groups spent similar time near the now-familiar social stimulus (Stranger 1) and a novel social stimulus (Stranger 2, Figure 9c; Table S1; main effect of Diet: $F(1, 52) = 4.157$, $p = 0.046$, partial $\omega^2 = 0.01$ (Small), 95% CI [0.00, 1.00]; post hoc: all comparisons $p > 0.05$); there was a similar outcome if data are presented as time spent in each chambers rather than time in sniff zone (data not shown). This is consistent with prior work that >17-mon old mice fail to show social novelty (Shoji & Miyakawa, 2019). Together, these data show that all four groups of C57BL6J 17-18-mon-old mice have normal sociability but reduced social novelty preference regardless of 33-GCR, CDDO-EA, or combination treatment.

The first day in the OF (Day 1) when the OF arena is novel, Veh/Sham and Veh/33-GCR mice moved a similar distance, whereas CDDO-EA/33-GCR mice covered on average 14.9% less distance than CDDO-EA/Sham mice (Figure 10a; Table S1; main effect IRR: $F(1, 47) = 4.673$, $p = 0.0358$, partial $\omega^2 = 0.07$ (Medium), 95% CI [0.00, 0.25]; post hoc: CDDO/Sham vs. CDDO/33-GCR: *f* $p = 0.030$, Cohen's $d = 0.874$ (Large), 95% CI [−1.64, −0.0333]). Mice

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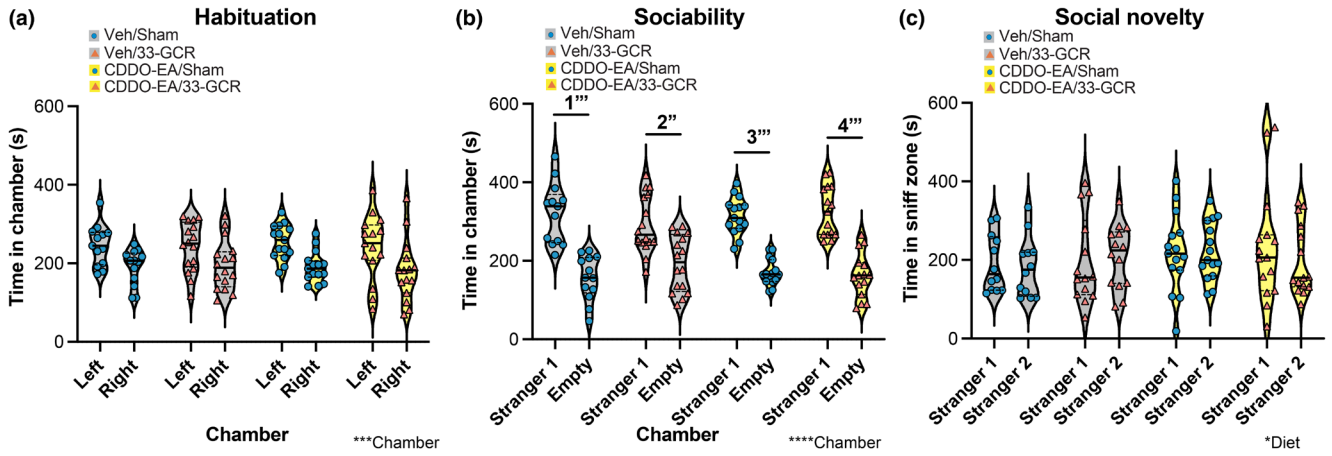


FIGURE 9 11.75 mon post-IRR, measures in the three-chamber sociability interaction (3-CSI) task are similar among groups. (a) 3-CSI habituation trial. (b) 3-CSI sociability trial reflecting time spent in the chamber containing a novel conspecific mouse in an enclosure (Stranger 1) and the chamber containing an empty enclosure (Empty). (c) 3-CSI social novelty trial reflecting time spent in the sniff zone around a now-familiar conspecific mouse in an enclosure (Stranger 1) and a novel conspecific mouse in the opposite enclosure (Stranger 2). Three-way ANOVA (Chamber × Diet × IRR). Main effects are denoted by $^{*}p<0.05$, $^{***}p<0.001$, or $^{****}p<0.0001$. 12–16 mice per group. IRR, irradiation. Veh, vehicle. Violin plots depict median (solid line) and 25% and 75% quartiles (dotted line). Complete data analysis details are provided in Table S1.

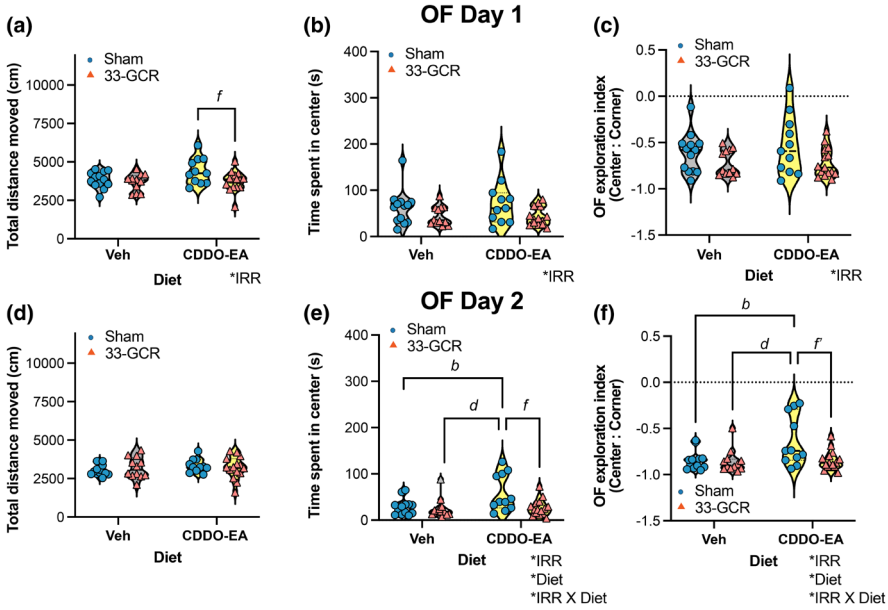


FIGURE 10 12 mon post-IRR, CDDO-EA/33-GCR mice showed less exploration in a novel, bright open field (OF) relative to CDDO-EA/Sham mice. In contrast, in a familiar OF CDDO-EA/Sham mice showed greater exploration relative to all other groups. (a–f) OF metrics on Day 1 (a–c) and Day 2 (d–f). (a) Day 1, distance moved. (b) Day 1, center time. (c) Day 1, OF exploration index. (d) Day 2, distance moved. (e) Day 2, center time. (f) Day 2, OF exploration index. Two-way ANOVA (Diet × IRR). Main effects are denoted by $^{*}p<0.05$. Bonferroni post-hoc multiple comparison analysis significance denotations between groups: Veh/Sham vs. CDDO-EA/Sham, *b* $p<0.05$. CDDO-EA/Sham vs. CDDO-EA/33-GCR, *d* $p<0.05$. CDDO-EA/Sham vs. CDDO-EA/33-GCR, *f*# $0.08<p<0.05$, *f* $p<0.05$. 11–15 mice per group IRR, irradiation. Veh, vehicle. Violin plots depict median (solid line) and 25% and 75% quartiles (dotted line). Complete data analysis details are provided in Table S1.

in the 33-GCR groups also spent less time in the center of the OF (Figure 10b; Table S1; main effect of IRR: $F(1, 47)=5.952$, $p=0.0185$, partial $\omega^2=0.09$ (Medium), 95% CI [0.00, 0.28]) and had a lower OF exploration index (Figure 10c; Table S1; main effect of IRR: $F(1, 47)=5.831$, $p=0.0197$, partial $\omega^2=0.09$ (Medium), 95% CI [0.00, 0.27]) vs. mice in the Sham groups, although there were no post-hoc differences.

On the second day of OF (Day 2), when the arena is now familiar, the total distance moved was similar among all four groups (Figure 10d; Table S1). However, two additional measures from OF Day 2 were

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informative. In regard to time in the center, CDDO-EA/Sham mice spent 104–129% more time in the center vs. Veh/Sham, Veh/33-GCR mice, and CDDO-EA/33-GCR mice (Figure 10e; Table S1; interaction of IRR×Diet: $F(1, 47)=4.130$, $p=0.0478$, partial $\omega^2=0.06$ (Medium), 95% CI [0.00, 0.23]; post-hoc: Veh/Sham vs. CDDO-EA/Sham: *b* $p=0.026$, Cohen's $d=1$ (Large), 95% CI [0.178, 1.84]; Veh/33-GCR vs. CDDO-EA/Sham: *d* $p=0.013$, Cohen's $d=1.04$ (Large), 95% CI [0.117, 1.82]; CDDO-EA/Sham vs. CDDO-EA/33-GCR: *f* $p=0.012$, Cohen's $d=1.09$ (Large), 95% CI [−1.96, −0.258]). OF exploration index followed the same pattern; CDDO-EA/Sham mice had a higher OF exploration index vs. Veh/Sham, Veh/33-GCR mice, and CDDO-EA/33-GCR mice (Figure 10f; Table S1; interaction of IRR×Diet: $F(1, 47)=4.958$, $p=0.0308$, partial $\omega^2=0.07$ (Medium), 95% CI [0.00, 0.26]; post-hoc: Veh/Sham vs. CDDO-EA/Sham: *b* $p=0.012$, Cohen's $d=0.209$ (Small), 95% CI [0.0609, 0.392]; Veh/33-GCR vs. CDDO-EA/Sham: *d* $p=0.010$, Cohen's $d=0.217$ (Small), 95% CI [0.0627, 0.394]; CDDO-EA/Sham vs. CDDO-EA/33-GCR: *f'* $p=0.009$, Cohen's $d=0.209$ (Small), 95% CI [0.0636, 0.39]). These data from OF Day 2 suggest that 11-12-mon post-IRR (17-18-mon-old mice) CDDO-EA induces an anti-anxiety-like effect which is negated by 33-GCR.

Habituation (adaptation to a novel environment) is important evolutionarily. We examined habituation and anxiety-relevant metrics 12 mon post-IRR by comparing OF on Day 1 and Day 2. For distance moved, two of the four groups moved less on Day 2 vs. Day 1, suggesting normal habituation: Veh/Sham mice (75.6% less on Day 2) and CDDO-EA/Sham mice (69% less on Day 2; Figure S2a; Table S1; main effect of Day: $F(1, 47)=45.71$, $p<0.0001$, partial $\omega^2=0.25$ (Large), 95% CI [0.06, 0.46]; post-hoc: Veh/Sham Day 1 vs. Day 2: *1"* $p=0.007$, Cohen's $d=1.65$ (Large), 95% CI [−2.66, −0.591]; CDDO-EA/Sham Day 1 vs. Day 2: *3‴*$p=0.0003$, Cohen's $d=1.55$ (Large), 95% CI [−2.35, −0.671]). Forfv time in the center, only Veh/Sham mice spent 53.7% less time in the center on Day 2 vs. Day 1 (Figure S2b; Table S1; main effect of Day: $F(1, 47)=27.88$, $p<0.0001$, partial $\omega^2=0.12$ (Medium), 95% CI [0.00, 0.32]; post-hoc: Veh/Sham Day 1 vs. Day 2: *1'* $p=0.001$, Cohen's $d=1.15$ (Large), 95% CI [−1.83, −0.35]). For OF exploration index, only Veh/Sham mice had a 37.6% lower OF exploration index on Day 2 vs. Day 1 (Figure S2c; Table S1; main effect of Day: $F(1, 47)=44.90$, $p<0.0001$, partial $\omega^2=0.14$ (Large), 95% CI [0.01, 0.35]; post-hoc: Veh/Sham Day 1 vs. Day 2: *1‴*$p<0.0001$, Cohen's $d=1.38$ (Large), 95% CI [−2.17, −0.571]). These data show that 11–12-mon post-IRR (17–18-mon-old mice), Veh/Sham mice and CDDO-EA/Sham mice habituate normally to a new environment, while Veh/33-GCR and CDDO-EA/33-GCR mice do not. In contrast, only Veh/Sham mice are also less anxious Day 2 vs. Day 1; CDDO-EA/Sham, Veh/33-GCR, and CDDO-EA/33-GCR mice show similar levels of anxiety-like behavior on both days.

Mice in all four groups faced with a novel and familiar object spent more time exploring the novel object (Figure 11a; Table S1; interaction of Object×Diet: $F(1, 56)=4.330$, $p=0.0420$, partial $\omega^2=0.02$ (Small), 95% CI [0.00, 0.14]; post-hoc: Veh/Sham, Familiar vs. Novel: *1‴*$p<0.0001$, Cohen's $d=1.57$ (Large), 95% CI [0.678, 2.56]; Veh/33-GCR, Familiar vs. Novel: *2"* $p<0.0001$, Cohen's $d=2.33$ (Large), 95% CI [1.17, 3.47]; CDDO-EA/Sham, Familiar vs. Novel: *3‴*$p<0.0001$, Cohen's $d=2.21$ (Large), 95% CI [1.32, 3.15]; and CDDO-EA/33-GCR, Familiar vs. Novel: *4‴*$p<0.0001$, Cohen's $d=1.96$ (Large), 95% CI [1.06, 2.67]). All groups also showed a similar discrimination index, favoring the novel object (Figure 11b; Table S1). Taken together, these data suggest that 12–13-mon post-IRR (18–19-mon-old mice), all four groups showed preference for a novel versus familiar object.

In the marble burying test, all four groups of mice buried a similar number of marbles (Figure 12a; Table S1). In the nestlet shredding

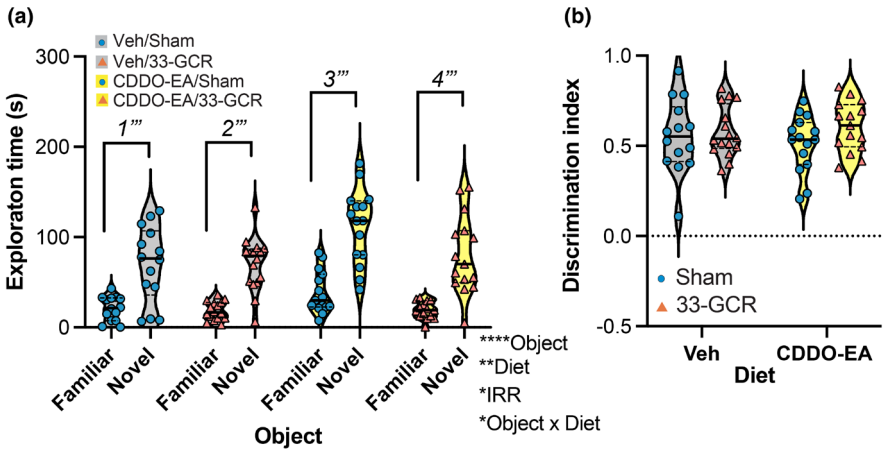


FIGURE 11 12.5 mon post-IRR, mice in all groups recognized novel objects, and they discriminated between novel and familiar objects to a similar extent. (a) Time spent exploring a novel versus familiar object. (b) Object discrimination index. (a) Three-way ANOVA (Object×Diet×IRR). Main effects and interactions are denoted by $^*p<0.05$, $^{****}p<0.0001$. Bonferroni post-hoc multiple comparison analysis significance denotations between objects for a given group: *1* = within Veh/Sham, *2* = within Veh/33-GCR; *3* = within CDDO-EA/Sham; *4* = within CDDO-EA/33-GCR with "=$p<0.001$ and "'=$p<0.0001$. (b) Two-way ANOVA (Diet×IRR). IRR, irradiation. s, seconds. Veh, vehicle. Violin plots depict median (solid line) and 25% and 75% quartiles (dotted line). 14–16 mice per group. Complete data analysis details are provided in Table S1.

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test, 33-GCR groups had marginally lower weight of intact nestlet versus Sham groups, but there was no post-hoc difference (Figure 12b; Table S1; main effect of IRR: $F(1, 56) = 6.094$, $p = 0.016$, partial $\omega^2 = 0.08$ (Medium), 95% CI [0.00, 0.25]). These data suggest equivalent performance in tests of compulsive behavior in all four groups when examined 12–13-mon post-IRR (18–19-mon-old-mice).

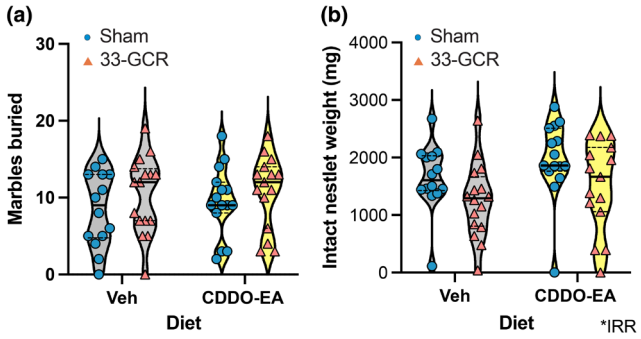


FIGURE 12 13 mon post-IRR, all groups of mice showed similar measures in tests of compulsive behavior. (a) Number of marbles buried. (b) Weight of intact nestlet. (a, b) Two-way ANOVA (Diet × IRR). 14–16 mice per group IRR, irradiation. Veh, vehicle. Violin plots depict median (solid line) and 25% and 75% quartiles (dotted line). Complete data analysis details are provided in Table S1.

After brains were collected (14.25 mon post-IRR) and processed for IHC, all mice showed evidence of adult-generated neurons, with DCX+ immature neurons in the subgranular zone of the dentate gyrus granule cell layer. DCX+ immature neurons had a characteristic pyramidal soma with processes of varying lengths emerging from the soma (Figure 13a). When DCX+ immature neuron number was examined as a single value per mouse, Veh/33-GCR mice had 27.4% fewer DCX+ immature neurons versus Veh/Sham mice. However, there was no difference in DCX+ immature neuron number in CDDO-EA/33-GCR and CDDO-EA/Sham mice (Figure 13b; Table S1; main effect of IRR: $F(1, 40) = 6.749$, $p = 0.0131$, partial $\omega^2 = 0.12$ (Medium), 95% CI [0.00, 0.33]; post-hoc: Veh/Sham vs. Veh/33-GCR, *a* $p = 0.0339$, Cohen's $d = 1.42$ (Large), 95% CI [−2.35, −0.407]). When DCX+ immature neuron number was examined in across the anterior–posterior axis of the dentate gyrus, there were fewer DCX+ immature neurons in Veh/33-GCR versus Veh/Sham mice at one anterior distance from bregma (bregma −2.02; Figure 13c; Table S1; main effect of Treatment: $F(3, 480) = 4.036$, $p = 0.0075$, partial $\omega^2 = 0.06$ (Medium), 95% CI [0.02, 0.11]; post-hoc at Bregma −2.02: Veh/Sham versus Veh/33-GCR: *a* $p = 0.0372$, Cohen's $d = 1.21$ (Large), 95% CI [−1.81, −0.488]) and fewer in CDDO-EA/33-GCR versus CDDO-EA/Sham mice at a distinct, more posterior distance from bregma (bregma −3.22; Figure 13c; Table S1; post-hoc at Bregma −3.22: CDDO/Sham versus CDDO/33-GCR: *f*

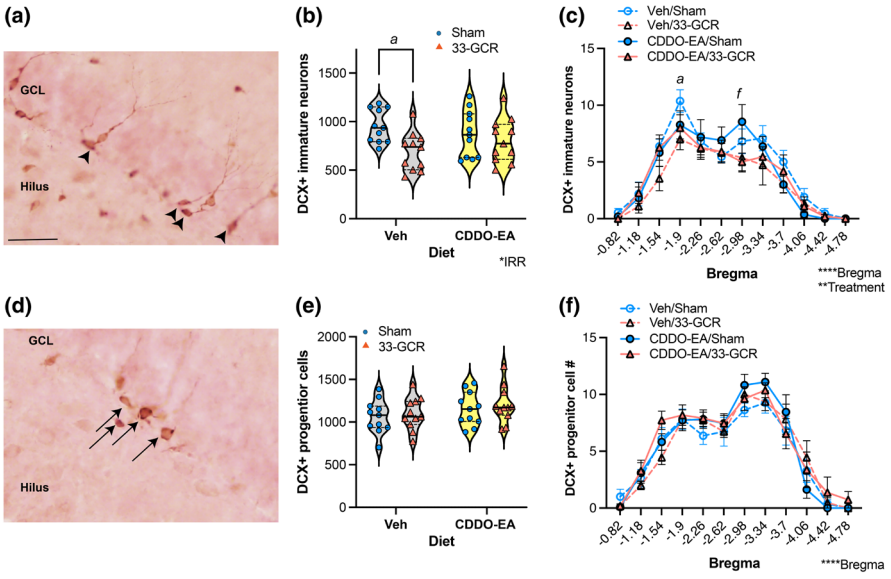


FIGURE 13 14.25 mon post-IRR, Veh/33-GCR mice had fewer doublecortin-immunoreactive (DCX+) immature neurons in the dentate gyrus (DG) subgranular zone (SGZ) vs. Veh/Sham mice, but all groups had a similar number of DCX+ progenitors. (a–c) DCX+ immature neurons and (d–f) DCX+ progenitor cells in the DG SGZ. (a) Representative photomicrograph of DCX+ immature neurons. Arrowheads indicate brown-stained DCX+ immature neuron soma in the SGZ at the border of the DG granule cell layer (GCL) and the DG hilus. (b) Number of DCX+ immature neurons represented as a single value per mouse. (c) Number of DCX+ immature neurons represented as a single value for each distance from bregma. (d) Representative photomicrograph of DCX+ progenitor cells. Arrows indicate brown-stained DCX+ progenitor cell soma in the SGZ. (e) Number of DCX+ progenitor cells represented as a single value per mouse. (f) Number of DCX+ progenitor cells represented as a single value for each distance from bregma. (b, e) Two-way ANOVA (Diet × IRR). Main effects and interactions are denoted by $^*p < 0.05$. Bonferroni post-hoc multiple comparison analysis significance denotations between groups: Veh/Sham versus Veh/33-GCR, *a* $p < 0.05$, CDDO/Sham versus CDDO/GCR, *f* $p < 0.05$. (c, f) Two-way ANOVA (Bregma × Treatment). Bonferroni post-hoc. +, immunoreactive. Scale bar in (a) = 100 μm, applies to (a, d). 11 mice per group. IRR, irradiation. Veh, vehicle. Violin plots (b, e) depict median (solid line) and 25% and 75% quartiles (dotted line). Complete data analysis details are provided in Table S1.

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$p = 0.0247$, Cohen's $d = 0.885$ (Large), 95% CI [−1.43, −0.111]). In addition to DCX+ immature neurons, all mice showed DCX+ progenitor cells which display an irregular soma and no processes emerging/seen in the Z plane of each cell (Figure 13d). Unlike DCX+ immature neuron number, there was no difference among the four groups of mice in DCX+ progenitor cell number either when considered as a single value per mouse (Figure 13e; Table S1) or examined across the anterior–posterior coronal axis (Figure 13f; Table S1; main effect of Bregma: $F(11, 480) = 71.85$, $p < 0.0001$, partial $\omega^2 = 0.60$ (Medium), 95% CI [0.53, 0.64]). These data show that 14.25-mon post-IRR (20.25-mon-old mice) 33-GCR leads to fewer DCX+ immature neurons but not progenitor cells relative to Sham mice.

4 | DISCUSSION

The estimated cumulative GCR dose on a mission to Mars varies. It depends on many things, including distance from solar minima/maxima, models used in dose estimation, instruments used in particle detection, mission architecture revisions, and spacecraft shielding modeling (Nelson, 2016). NASA has named a range of GCR doses relevant for radiobiology experiments, from 12.5 to 75 cGy (Simonsen et al., 2020). This range covers doses experienced during exploratory class missions under consideration by NASA. NASA includes the "higher" exposure of 75 cGy—the dose used in the present study—to establish a dose response curve above expected mission exposures. For example, our data shed light on the effect GCR had on cognition when ca. 1570 days are spent in deep space. Including the 75 cGy dose also helps NASA in the evaluation of how mixed-field quality effects inform permissible exposure limits. In fact, exposures up to 1.5 Gy may be warranted in some cases where clinically significant endpoints are not evident in animal model systems at lower doses, such as the determination of quality effects on the cardiovascular system (Simonsen et al., 2020), or to attempt to factor dose contributions from solar particle events, which range widely depending on the solar cycle, intensity and direction of a solar particle event. Of note, the 75 cGy dose used here far exceeds that used for clinical imaging. Also, the differences in dose response and radiation properties between x- and gamma-rays used in clinical imaging vs. the high-energy charged particles that comprise GCR make radiobiological comparisons on the basis of dose impossible; this highlights the need for studies using whole-body particle radiation. Thus, the whole-body charged-particle radiation dose used here—75 cGy—is a realistic and important dose to use in animal model experiments to understand the impact that long duration space mission, such as to Mars, has on performance in a variety of cognitive domains.

In our work, mice underwent testing for performance in spaceflight-relevant behavioral and cognitive tests 2 to 13.25 mon after administration of CDDO-EA or Veh chow and exposure to the Mars-relevant dose of 75 cGy 33-GCR or Sham IRR. Only a few studies have looked at rodent behavioral performance at such long time points post-IRR with mixed beams at Mars-relevant doses: >5 mon in mouse (Kiffer et al., 2021; Rienecker et al., 2023) and 12-mon post-IRR in rat (Raber et al., 2021). The only work that has looked at behavioral performance after 75 cGy of 33-GCR has looked in male mice (Kiffer et al., 2021), but not female mice. No behavior studies have looked at mouse performance >6 mon post-IRR with mixed beams. This is a major knowledge gap, as space radiation-induced changes in behavioral performance are highly dependent on time post-IRR (Miry et al., 2021). We also included social measures, as work the effect of Mars-relevant doses of charged particle radiation on social behaviors has been performed in slightly younger mice (7.5- to 15-mon-old, 1.5- to 9-mon post-IRR; Jones et al., 2019; Kiffer et al. 2019a; Krukowski et al. 2018a; Krukowski et al. 2018b; Mange et al., 2018). We felt it important to test social behaviors of mice at a slightly older age (17- to 18-mon-old, ca. 11.75 mon post-IRR). One might question our testing of social novelty/social memory at this older age since older mice (17- and 23-mon-old at testing) maintain sociability but are deficient in social novelty/social memory (Shoji & Miyakawa, 2019). However, behavioral results after radiation are sometimes surprising. For example, we previously predicted mice exposed to charged-particle radiation would perform worse on behavioral pattern separation tasks, but they actually perform better (Soler et al., 2021; Whoolery et al., 2020). Therefore, we felt it important to directly assess social novelty in our study. In general, the results presented here on behavioral and social performance up to 13.25 mon post-IRR will be useful for the field and for NASA as they establish astronaut risk modeling.

In fact, the data we present are multi-faceted and provide a long-term perspective of how exposure to 33-GCR influences cognition and behavior in female mice, and how administration of the antioxidant CDDO-EA counters 33-GCR-induced changes. Below we discuss the main results of this longitudinal study and integrate these new findings into the existing space radiation and cognition literature.

In the touchscreen platform, one of the most informative cognitive changes we report is that Veh/33-GCR mice are slower to acquire a simple stimulus–response association when tested 9-mon post-IRR (Figure 7c; large effect size). These mice can learn the task within 30 min (data not shown) but appear simply slower in reaching the time cut-off often used for this task (finishing within 15 min). Supporting their slower completion, when the entire 30 min of each session is considered, Veh/33-GCR mice have longer sessions (Figure 7d; large effect size) and longer latencies to retrieve the reward (Figure 7f; large effect size). Veh/33-GCR mice have, however, normal latency to make a correct touch (Figure 7e) and near-perfect accuracy (Figure 7g). One way to interpret these data is that in this simple stimulus–response association task, Veh/33-GCR mice know what to do and they do it correctly, but they take a longer time to retrieve the reward. This relative slowness could be due to motoric issues, motivation issues, or cognitive issues. The latter is unlikely, as the near-perfect accuracy suggests the relative slowness is not due to cognitive dysfunction. Given that Veh/33-GCR mice make fewer beam breaks and have longer session lengths and latencies to collect the reward, we believe Veh/33-GCR mice have motoric issues, motivation issues, attention issues, or a combination. In regard

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to attention, prior work shows male mice exposed to a slightly lower dose of 33-GCR (40–49 cGy vs. our 75 cGy) have worse attentional processes when tested 6 mon post-IRR (Desai et al., 2023). However, other prior work in the simple stimulus–response learning task suggests that one of the metrics (correct touch latency) reflects attentional processes (Swan et al., 2014), and radiation does not change correct touch latency in our mice (Figure 7e). Therefore, we do not favor the interpretation that at this time point 33-GCR mice have disrupted attention. Other work in male mice showed no change in reward sensitivity in male mice examined 6 mon post-IRR (Desai et al., 2023), which could be considered to be reflective of reward motivation. It is possible that there are motivation deficits in our mice at this time point because of a higher dose of 33-GCR used here (75 cGy) relative to prior work (40-49 cGy vs. our 75 cGy; Desai et al., 2023), or that motivation effects are different at distinct time points post-IRR. Future work is needed to dissect if the changes in Veh/33-GCR mice we show here are due to an impact on motor or motivational processes or a combination. On a related note, we also report administration of CDDO-EA at the time of IRR (9 mon prior) prevented the 33-GCR-induced decrease in reward collection latency (Figure 7f), perhaps normalizing reward motivation in these female mice. Although translating motivational findings from mice to humans presents challenges, a loss of motivation in humans is associated with anhedonic behaviors such as a loss of arousal, loss of health-oriented behaviors, and overall well-being. Of note, Mars mission-relevant doses of particle radiation have demonstrated deleterious effects on the dopaminergic system, which is tightly linked to motivational processes (Davis et al., 2014; Davis et al., 2015; Morris et al., 2022). Deficits to motivational systems and/or motor processes present serious challenges to Mars-bound astronauts, and work with animal models merits further evaluation both from radiation risk assessment and astronaut fitness screening perspectives.

Another point of discussion about the touchscreen platform cognitive data we present is the lack of change seen in female mice in behavioral pattern separation (Small separation, Figure 4), a mission-relevant task reliant on the integrity of the hippocampal dentate gyrus. This is in contrast to prior work showing male and female C57BL/6J mice exposed to a single high-energy particle regimen (i.e. ^{56}Fe, ca. 20 cGy total) performed *better* than Sham IRR mice in a behavioral pattern separation touchscreen task (LDR) 6 to 8 mon post-IRR (Soler et al., 2021; Whoolery et al., 2020). In our current work, female C57BL/6J mice were examined 3.5 to 9 mon after 33-GCR and showed no gross difference in LDR performance (Figure 4). Thus, we conclude that this dose (ca. 75 cGy total in 1 day) of whole-body 33-GCR does not influence behavioral pattern separation assessed via LDR, at least not at this time examined post-IRR. In contrast, other work shows that female C57BL/6J mice that were exposed to 33-GCR either acutely (40 cGy over 2 h) or repeatedly (50 cGy over 24 days) had *worse* behavioral pattern separation relative to control female mice, as assessed in the object in updated location test (OUL; Alaghband et al., 2023). This prior work and our current 33-GCR work cannot be directly compared, given the different dose-rates and that the distinct behavioral pattern separation tests used (LDR in ours, OUL in theirs) may have different levels of sensitivity to dysfunction. However, clearly more work is warranted on the impact of 33-GCR on behavioral pattern separation in mice. Our present negative LDR finding aside, we also present two interesting CDDO-EA LDR findings. First, in measures of discrimination (Large separation, Figure 4b,f,j), female mice that received CDDO-EA at the time of 33-GCR had *better* dentate gyrus-dependent discrimination vs. CDDO-EA/Sham IRR mice, as shown by greater accuracy and shorter session length. In fact, this leads to the second finding: female mice that received CDDO-EA had *better* behavioral pattern separation (Small separation, Figure 4d,h,l) than Veh mice, regardless of IRR (33-GCR or Sham IRR). It is tempting to conclude that CDDO-EA may improve cognition in the long-term. However, much more work would be needed prior to making such a conclusion, including addressing the possibility that prior behavior testing influenced performance. To this end, one approach would be to test parallel groups of mice in distinct cognitive tasks at the same time point after CDDO-EA administration.

The arena-based analyses we performed also provided several points for discussion, the first being the lack of difference we report between female 33-GCR and Sham IRR mice in OF locomotion (Figure 10). A few studies have established that single particle exposure leads to fine motor deficits in rats (Blackwell et al., 2021; Blackwell et al., 2022; Blackwell et al., 2023), but differences in gross locomotion are scarce in the literature with studies reporting that post-IRR OF locomotion is increased (Boutros et al., 2021; Kokhan et al., 2019; Liu et al., 2019; Sorokina et al., 2020), decreased (Allen et al., 2015; Holshouser et al., 2020), or unchanged (Impey et al., 2023). Time post-IRR and dose may be important in seeing 5-ion (1H, 2×1000 MeV/n; 4He, 400 MeV/n; 16O, 400 MeV/n; 28Si, 600 MeV/n) GCR-induced changes in locomotion (Puukila et al., 2023), although there are conflicting results (Boutros et al., 2021). The potential impact of time post-IRR on locomotion was one reason we examined this metric at several time points post-IRR. Prior work exposing mice to 33-GCR have not reported differences in locomotion, neither in male mice recorded in activity chambers (75 cGy) nor in male or female mice recorded in an OF (acute 40 cGy or chronic 50 cGy across 4 weeks; Kiffer et al., 2021; Schaeffer et al., 2022). In keeping with this recent 33-GCR literature, we also report no radiation-dependent differences in gross locomotion in female mice receiving 75 cGy 33-GCR in either the activity chambers (Figure 8) or in an OF (Figure 10). Curiously, when the environment was novel (in the activity chambers and OF arena on Day 1), female CDDO-EA/33-GCR mice spend less time moving than CDDO-EA/Sham mice. However, when the environment was familiar (in the touchscreen chambers for habituation for Acq and OF arena on Day 2), there was no difference in movement between CDDO-EA/Sham and CDDO-EA/33-GCR mice. This implies the effect of CDDO-EA on locomotor activity is more likely to be seen in a novel environment, a fact that may be useful in future study of this potential countermeasure. This may also be the result of potential increase in impulsivity of 33-GCR on motivation in mice given

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CDDO-EA, a mission-relevant domain that should be evaluated further.

There are many other arena-based tests in which we report no difference after 33-GCR and/or CDDO-EA, including sociability, social novelty, anxiety/exploratory drive, habituation learning, object recognition memory, and compulsive-like behavior. Some of these negative findings are useful to mention here. For example, we find sociability was intact regardless of radiation or diet; we did not observe preference for social novelty in any group, likely due to the advanced age at the time of 3-CSI (16 mon-of-age). This stands in contrast to prior work done in 11.5-mon-old male mice, where most groups (Veh/Sham, Veh/33-GCR, CDDO-EA/Sham) had intact sociability except for mice receiving CDDO-EA and 33-GCR, and only Sham/Veh mice had intact social novelty while Veh/33-GCR, CDDO-EA/Sham, and CDDO-EA/33-GCR mice did not (Kiffer et al., 2021). One striking aspect of our current female and prior male 33-GCR work is the lack of effect on object recognition memory. The most commonly reported behavioral deficit after whole-body charged-particle radiation is in novel object recognition (NOR) (Cacao & Cucinotta, 2019). However, prior work with female mice also shows intact NOR 3.5 mon after acute (40 cGy) or chronic (50 cGy) 33-GCR; notably, male mice in that study had disrupted NOR (Alaghband et al., 2023). Thus, future work should test if 33-GCR-induced deficits in NOR are indeed sex-specific.

In addition to assessing the impact of 33-GCR and/or CDDO-EA on behavior and cognition, we also assessed the impact on the number of DCX+ cells in the dentate gyrus SGZ. In rodents, DCX-immunoreactivity plus morphology has been considered a "gold-standard" approach for the detection of immature neurons and thus provides a cellularindex relevant to adult neurogenesis (Flor-García et al., 2020; Plümpe et al., 2006). Radiation exposure has long been associated with detrimental effects on indices of hippocampal dentate gyrus neurogenesis (McNerlin et al., 2022). Monoenergetic single particle radiation paradigms typically decrease indices of dentate gyrus neurogenesis (Casadesus et al., 2005; DeCarolis et al., 2014; Rivera et al., 2013; Rola et al., 2004; Rola et al., 2005; Rola et al., 2008; Simmons et al., 2022; Whoolery et al., 2017; Whoolery et al., 2020; Zanni et al., 2018). However, our data presented here are the first to assess if multiparticle exposure (33-GCR) at a mission-relevant dose can change indices relevant to hippocampal neurogenesis (DCX+ immature neurons and cells) compared to Sham IR. We find female mice 14.25 mon post-33-GCR have fewer DCX+ immature neurons than Sham IRR mice (large effect size); DCX+ progenitor cell number is unchanged. Dentate gyrus neurogenesis has been considered crucial for behavioral pattern separation in young adult rodents (Clelland et al., 2009; Nakashiba et al., 2012; Sahay et al., 2011) although it does not always predict pattern separation ability, particularly in aged mice (Cès et al., 2018; Saxe et al., 2007; Whoolery et al., 2020). Here we find no difference in behavioral pattern separation between Veh/Sham mice and Veh/33-GCR mice. However, we do not suggest a relationship between any behavioral outcome in the present study and the indices of neurogenesis we measure because they were assessed at distinct times post-IRR; behavioral pattern separation was tested between 3.5 and 8.5 mon post-IRR, and indices relevant to neurogenesis were examined 14.25 mon post-IRR. The process of neurogenesis is dynamic and age-dependent, and our DCX+ cell number data capture this neurogenesis index at just one point in time. Future studies ideally would be able to probe a causative role for neurogenesis, for example, using available mouse models to inducibly increase or decrease neurogenesis prior to IRR and seeing how pattern separation is changed post-IRR.

There are at least three limitations of the present study. First, the dose-rate of the radiation delivered in our study is much higher than what would occur during space travel. Our mice were exposed to a whole-body acute dose of 75 cGy 33-GCR over 1.5 h, a dose-rate determined by best practices given the BNL NSRL accelerator constraints (Huff et al., 2023; Norbury et al., 2016; Simonsen et al., 2020); in spaceflight, there is a constant 0.5 mGy/day exposure behind standard spacecraft shielding during solar maximum (Zeitlin et al., 2013). In fact, it remains an open question if exposure to a GCR spectrum will reveal dose-rate effects, that is, if delivery of a GCR spectrum over a short vs. long period of time results in different effects. One study compared delivery of GCR over a short period of time (acute) vs. longer period of time (chronic), and reported decreased rodent performance in various behavioral tests after both dose-rates with one not more detrimental than the other. However, the study used an unbalanced dose design, so it is not possible to distinguish a dose-rate effect from a dose effect (Alaghband et al., 2023). Clearly more specific work on dose-rate effects is needed. A second limitation is that the potential countermeasure CDDO-EA here was delivered in chow to mice over a few days (before, during, and 1 day after IRR). This timing of the CDDO-EA dosing was by design; greatest protection from IRR-induced oxidative damage likely will result from short-term saturated plasma concentration (Kim et al., 2012; Wang et al., 2014). In deep space, however, the fluence of charged particles is constant, and thus future use of CDDO-EA in space travel would likely require a longer dosing period which may exert different pharmacological protective effects than described here. A final limitation is that we do not include a control, standard chow group; mice received either Veh chow or CDDO-EA chow for 5 days before, during, and 1 day after IRR. This is because we (and prior researchers) assumed the low amounts of the components of the Veh chow (1.25% EtOH, 3.75% Neobee Oil) rendered them benign to physiological processes. We remain open to the idea that the Veh chow itself affects the outcomes we measured here, something that could be tested in future studies. We recommend that future work assess GCR risks at varying doses and dose rates, along with chronic administration of potential countermeasures, and consider using a control, standard chow group.

In summary, mature female C57BL/6J mice exposed to 33-GCR had worse stimulus–response learning and fewer DCX+ immature dentate gyrus neurons relative to control mice over a year post-IRR, yet they had normal performance in tests for behavioral pattern separation, anxiety-like behavior, exploration, object recognition, and habituation. As for the potential countermeasure CDDO-EA, one radiation effect/CDDO-EA effect emerged: CDDO-EA prevented

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33-GCR-induced deficit in stimulus response learning. There were also CDDO-EA effects: CDDO-EA/Sham and CDDO-EA/33-GCR mice had better behavioral pattern separation vs. their respective control groups (Veh/Sham, Veh/33-GCR), and CDDO-EA/33-GCR mice had better cognitive flexibility (reversal number) vs. Veh/33-GCR. Our study implies space radiation is a risk to a female crew's longitudinal mission-relevant cognitive processes and CDDO-EA is a potential dietary countermeasure for space-radiation CNS risks.

AUTHOR CONTRIBUTIONS

S. Yun: Conceptualization; investigation; funding acquisition; writing – original draft; methodology; validation; visualization; writing – review and editing; formal analysis; project administration; data curation; supervision; resources. **F. C. Kiffer:** Investigation; methodology; funding acquisition; writing – original draft; visualization; writing – review and editing; conceptualization; formal analysis; project administration; supervision; data curation. **G. L. Bancroft:** Data curation; formal analysis; investigation. **C. S. Guzman:** Data curation; investigation; methodology. **I. Soler:** Investigation; data curation; methodology. **H. A. Haas:** Data curation; formal analysis; investigation; validation. **R. Shi:** Software; investigation; validation. **R. Patel:** Data curation; formal analysis; software; validation. **J. Lara-Jiménez:** Formal analysis; validation. **P. L. Kumar:** Investigation; validation. **F. H. Tran:** Investigation. **K. J. Ahn:** Investigation. **Y. Rong:** Software; writing – review and editing. **K. Luitel:** Investigation; methodology. **J. W. Shay:** Conceptualization; funding acquisition; methodology; project administration; resources; supervision; writing – review and editing. **A. J. Eisch:** Conceptualization; formal analysis; funding acquisition; resources; supervision; writing – original draft; writing – review and editing.

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CONFLICT OF INTEREST STATEMENT

The authors have no conflict of interests to declare.

PEER REVIEW

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DATA AVAILABILITY STATEMENT

Data are available on reasonable written request and will be available after publication at Ames Life Sciences Data Archive (ALSDA). A preprint of this article was posted on BioRxiv on April 15th, 2024: https://www.biorxiv.org/content/10.1101/2024.04.12.588768v1.

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